User Manual

Virtual Vehicle Validation Model (V3M) Tool / Platform

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Author: TCS Data Scientists:

(Dr Arindam Ghosh and Alexander Gyetvai)

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1. Introduction

1.1. Aims of the V3M Project

This solutions goal is to save time and cost of resources, to run continuous simulations without a person in the simulator and to rapidly and robustly validate changes to developmental software in the digital twin. Traditional real-world testing in Formula E (*The Official Home of the ABB FIA Formula E World Championship*, 2025) provides invaluable benchmarks, but it comes with significant drawbacks. On-track testing puts drivers and cars at risk of accidents, wear, and damage. Regulations also limit the number of sensors fitted to the car, restricting the depth of data teams can gather. Within motorsports are strict regulations set by the regulatory body, the Fédération Internationale de L'automobile (FiA) (*https://www.fia.com/regulation/category/109*, 2025). In this environment, every decision must maximise performance while minimising risk and cost. Therefore, utilisation of Driver-in-the-Loop (DiL) Simulators, and digital twins provide a realistic benchmark within a virtual environment in support of the Jaguar TCS Racing Team's objectives of winning World Championships (*Jaguar TCS Racing Season 11 | Formula E Team*, 2025).

1.2. Purpose of the V3M User Manual

This user manual provides comprehensive guidance on using the Virtual Vehicle Validation (V3M) application, a simulation interface for testing virtual vehicle dynamics in real time. The V3M allows users to configure simulation parameters, initiate and control simulation sessions, and visualise results from a simulator through UDP socket communication.

1.3. Audience

This manual is intended for simulation engineers, race engineers, developers, and test operators of Jaguar TCS Racing who work with Driver-in-the-Loop (DiL) or development (Offline) simulation environments. No prior knowledge of the V3M is required to follow this manual.

2. System Requirements

2.1. Minimum System Requirements

- **Operating System**: Windows 10
- **Python Version**: 3.11.5
- **Graphics**: Required for matplotlib visualizations
- **Network**: Local or network UDP communication support

2.2. Dependencies

Ensure the following Python packages are installed *not required for executable*:

Modules:

- Offline_Balance_Client.py
- steer_control.py,
- pedal_control.py,
- plot_graphs.py (for developmental work)
- V3M_Ext_Driver.py

A requirements.txt file is located in the repository for ease of installation

3. Installation & Access

3.1. Installing the Application

The V3M application is packaged as a standalone executable file for ease of access. No Python installation is required on the user's machine.

3.2. To clone the repository:

3.2.1. Clone the DataScience_ExtDriverModel Repository using

- HTTPS: https://catapult@dev.azure.com/catapult/Vehicle%20Performance/_git/DataScience_ExtDriverModel
- SSH: git@ssh.dev.azure.com:v3/catapult/Vehicle%20Performance/DataScience_ExtDriverExtDr
- VS Code: Select "Clone in VS Code"

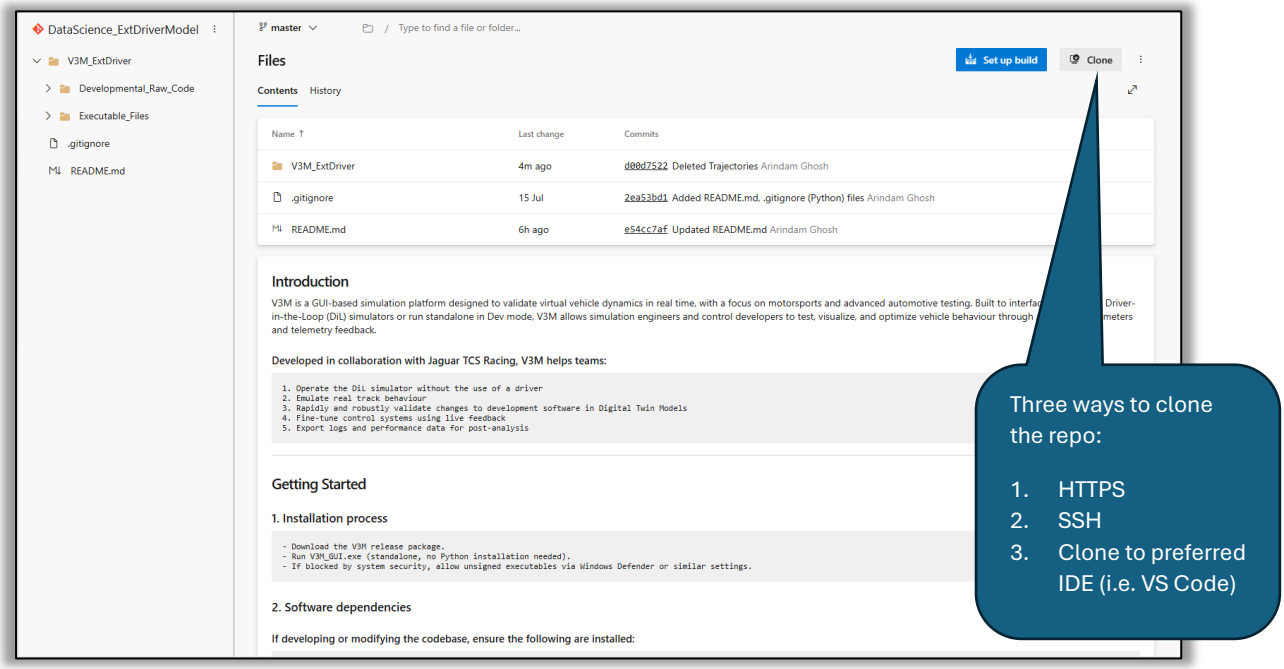


Figure 1: Overview of the V3M ExtDriver Repository

3.2.2. To run the Application:

- The V3M_ExtDriver folder consists of two subfolders:
  - Developmental_Raw_Code
  - Executable_Files
- To launch the complete version of the V3M, navigate to the "Executable_Files" subfolder.
- "Developmental_Raw_Code" is strictly for developers.

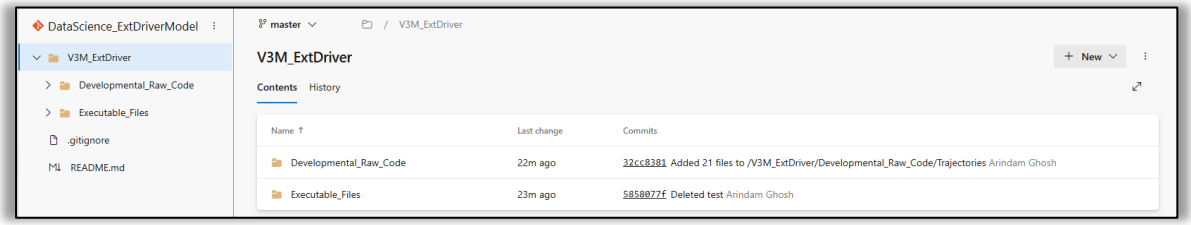


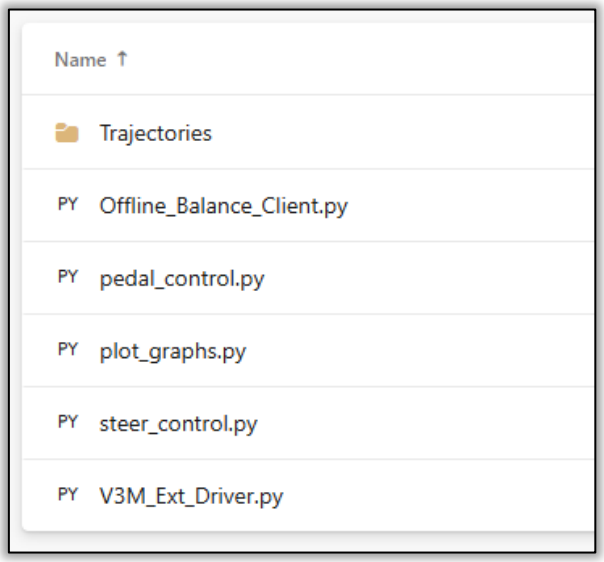
Figure 2: V3M_ExtDriver Folders

- The "Executable_Files" folder consists of two items: *V3M_ExtDriver.exe* and the *Trajectories* folder
  - *V3M_ExtDriver.exe* will launch the V3M Application
  - *Trajectories* folder – consists of the track data in .csv format
  - Double-click *V3M_ExtDriver.exe* to launch the interface.
  - The *V3M_ExtDriver.exe,* "Trajectories" folder and User Manual is also found with the file path: "\\cust-fs02\jaguarracing\Software\V3M"

Ensure your system allows execution of unsigned executables if prompted by security settings.

3.2.3. To use the Raw Development Code:

- Navigate to the "Developmental_Raw_Code" folder
- The "Developmental_Raw_Code" folder consists of 5 python files:
  - *Offline_Balance_Client.py* – Used for offline testing
  - *pedal_control.py* – Used for controlling V3M velocity
  - *plot_graphs.py* – Used for applying graph plotting logic
  - *steer_control.py* – Used for controlling V3M steering
  - *V3M_Ext_Driver.py* – Main file with UI and Simulation Loop logic
  - *Requirement.txt* – Contains all the required python packages
- There is also the *Trajectories* folder with track data in .csv format
- Download the latest version of Anaconda from https://www.anaconda.com/download/success
- Create an environment using the Anaconda Powershell Prompt:
  conda create -n V3M_env python=3.11.5
- Activate the environment using *conda activate V3M_env* in your development IDE
- Pip install the Requirements file using *pip install -r /path/to/requirements.txt*
- Run the *V3M_Ext_Driver.py* script to test run any development changes in the code.
- To create a .exe file:
  - Within the terminal of the IDE where the "Developmental_Raw_Code" is saved and *V3M_env* is activated, execute the command:
    - *pyinstaller --onefile "/path/to/V3M_Ext_Driver.py"*
  - This creates a "dist" folder with "*V3M_Ext_Driver.exe"* inside
  - Create a new folder and add "*V3M_Ext_Driver.exe"* and the "Trajectories" folder inside, in the same file structure as the "Executable_Files" folder to launch the application


Name
Trajectories
PY Offline_Balance_Client.py
PY pedal_control.py
PY plot_graphs.py
PY steer_control.py
PY V3M_Ext_Driver.py

4. User Interface Overview

The application interface is divided into the following key sections:

- IP Address and Port Settings
- Connection/Simulation Controls (Listen, Start, Stop, Abort, Reset)
- Parameter Selection Input Fields (General Parameters)
- Force Driver Input Fields
- Steering Control Parameters Input Fields
- Pedal Control Parameters Input Fields
- Torque Control Parameter Input Fields
- Show Graphs and Console Output

Each parameter, if hovered over with the curser, populates a tooltip window explaining the parameter. For a detailed overview of the parameter fields, please see the Appendix (under Section 16.1).

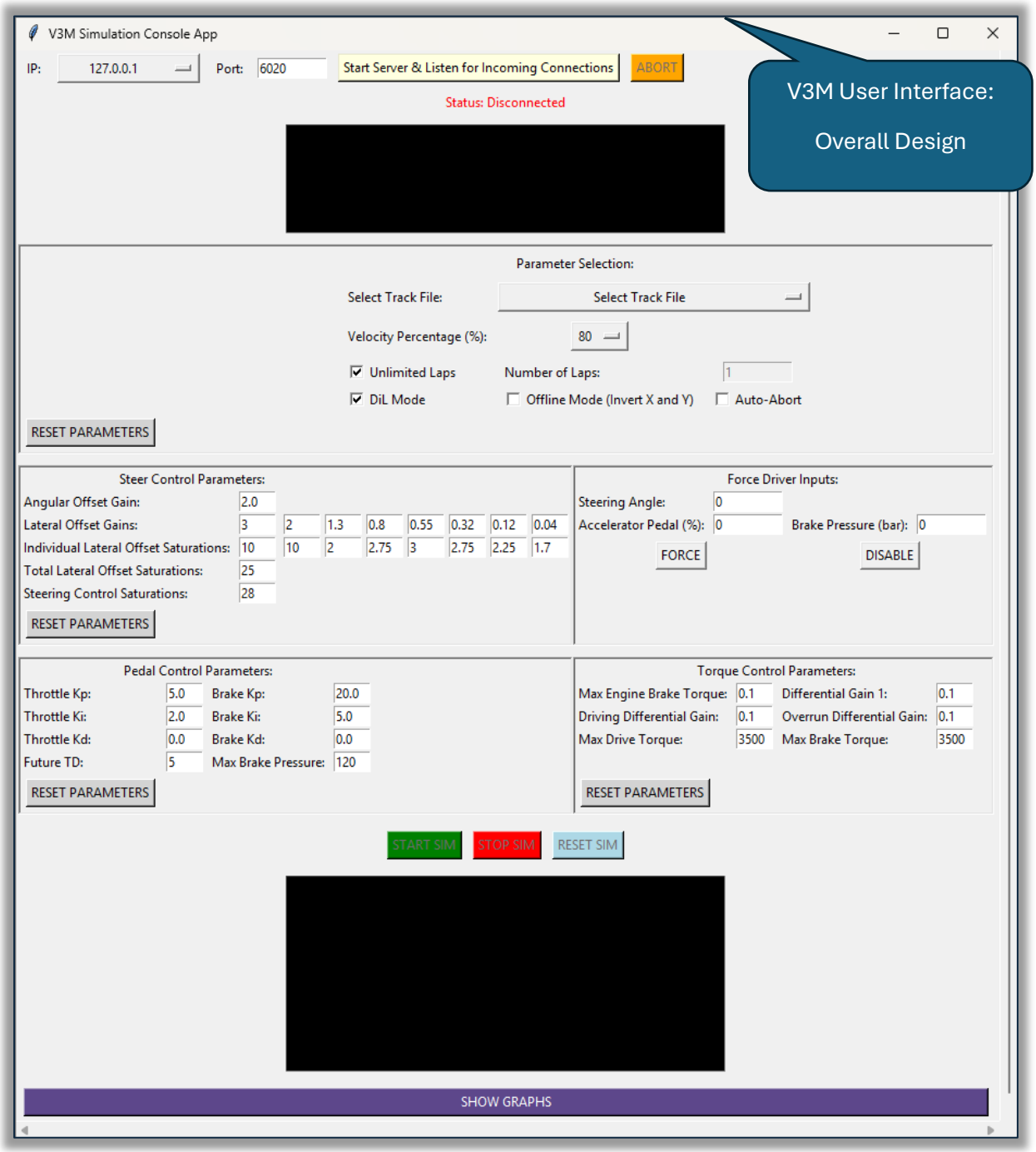


Figure 3: User Interface - Overall Design

5. Exporting Trajectory Data

To use BME input data, export from ATLAS your chosen lap with the following signals:

- sLapDistance
- vCar
- xCarX
- xCarY
- xCarZ
- aYaw

Save your data as a **.csv file** to the folder "Trajectories". These files will populate the "Select Tracks" dropdown menu. Please note that ATLAS 10 did not offer lap exporting, so please use ATLAS 9 to acquire this information (*Overview - ATLAS Product Suite*, 2025).

Please see below for correct structure.

	A	B	C	D	E	F	G
1		sLapDistance	vCar	xCarX	xCarY	xCarZ	aYaw
2	00:00.0	1.34	185.73	-25.93	-209.36	9.47	-148.04
3	00:00.1	3.93	186.61	-23.73	-207.99	9.43	-148.02
4	00:00.1	6.52	187.51	-21.53	-206.62	9.39	-147.99
5	00:00.2	9.13	188.39	-19.31	-205.23	9.35	-147.97
6	00:00.2	11.76	189.26	-17.09	-203.84	9.31	-147.94
7	00:00.3	14.39	190.12	-14.86	-202.45	9.27	-147.91
8	00:00.3	17.04	190.98	-12.61	-201.04	9.23	-147.88
9	00:00.4	19.7	191.83	-10.36	-199.63	9.2	-147.85
10	00:00.4	22.37	192.67	-8.1	-198.21	9.16	-147.82
11	00:00.5	25.05	193.5	-5.83	-196.78	9.12	-147.78
12	00:00.5	27.74	194.33	-3.55	-195.35	9.09	-147.74
13	00:00.6	30.45	195.11	-1.26	-193.91	9.05	-147.69
14	00:00.6	33.16	195.92	1.03	-192.46	9.01	-147.63
15	00:00.7	35.89	196.72	3.34	-191	8.97	-147.57
16	00:00.7	38.63	197.51	5.65	-189.54	8.93	-147.52
17	00:00.8	41.37	198.32	7.97	-188.06	8.88	-147.47
18	00:00.8	44.13	199.09	10.3	-186.58	8.84	-147.45
19	00:00.9	46.9	199.86	12.63	-185.1	8.8	-147.43
20	00:00.9	49.69	200.62	14.98	-183.6	8.75	-147.43

Table 1: Trajectory Input Data Structure as .csv file

6. Connecting to the Simulator

6.1. IP Address and Port Input

To establish communication with the simulator, you must provide:

- **IP Address**: x.x.x.x (IP Address to be provided by the DiL Team)
- **Port Number**: xxxx (Port Number to be provided by the DiL Team)

Ensure these values match the simulator settings before starting the server.

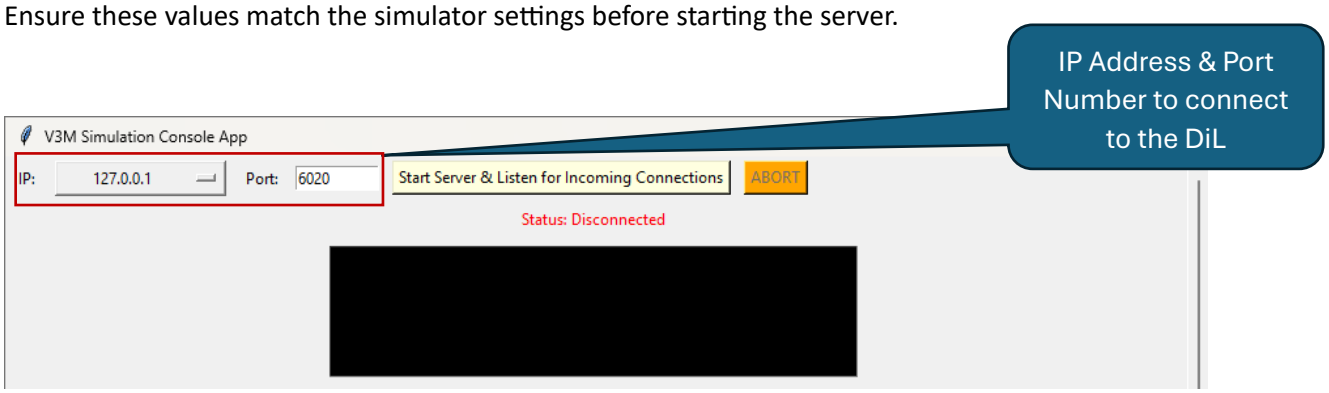


Figure 4: Connecting to the Simulator - IP Address & Port Number

6.2. Start Server & Listen Button

- Press **"Start Server & Listen for Incoming Connections"** to begin listening for the simulator. When pressed, the Status will turn to blue and say "Listening".

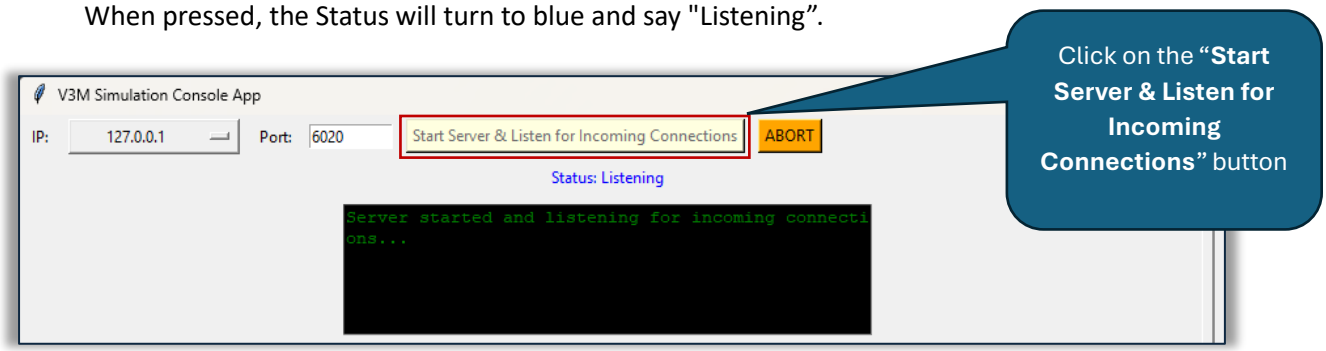


Figure 5: Start Server and Listen for Incoming Connections

- Once the DIL simulator is launched and attempts to connect,

  the GUI will automatically establish the connection.

- Console output will confirm a successful handshake.
- If successful, the Status will say "Connected"

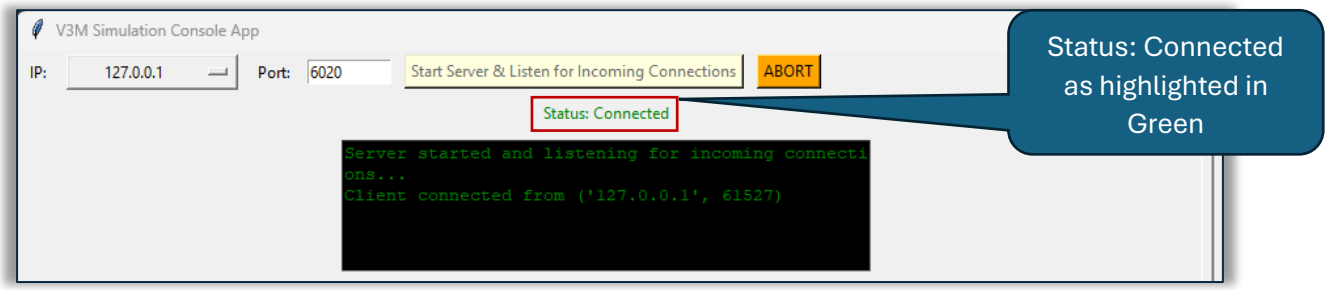


Figure 6: Connecting to the Client

7. V3M Parameters

- The V3M parameters are subdivided into different categories

7.1 Parameter Selection

- These parameters determine the general capabilities of the V3M. Here you may modify:
    - Track Selection
    - Velocity Percentage
    - Number of Laps/Unlimited laps
    - V3M Modes (DiL/Offline Mode)
    - Auto-Abort initiated

7.2 Steer Control Parameters

- These parameters determine the how the V3M calculates the steering input using:
    - Angular Offset Gains
    - Lateral Offset Gains
    - Saturations for each individual Lateral Offset Gain
    - Total Lateral Offset Saturation
    - Steering Control Saturation

7.3 Pedal Control Parameters

- These parameters determine the how the V3M calculates the accelerator pedal percentage and brake pressure using velocity error against the target velocity profile
    - Throttle PID Gain Values
    - Brake PID Values
    - Future Target Distance
    - Max Brake Pressure

7.4 Torque Control Parameters

- These parameters serve to assist in torque calculations within offline testing (where drive/brake torques are required instead of pedal controls).
    - Max Engine Brake Torque
    - Differential Gains
    - Max Drive/Brake Torques

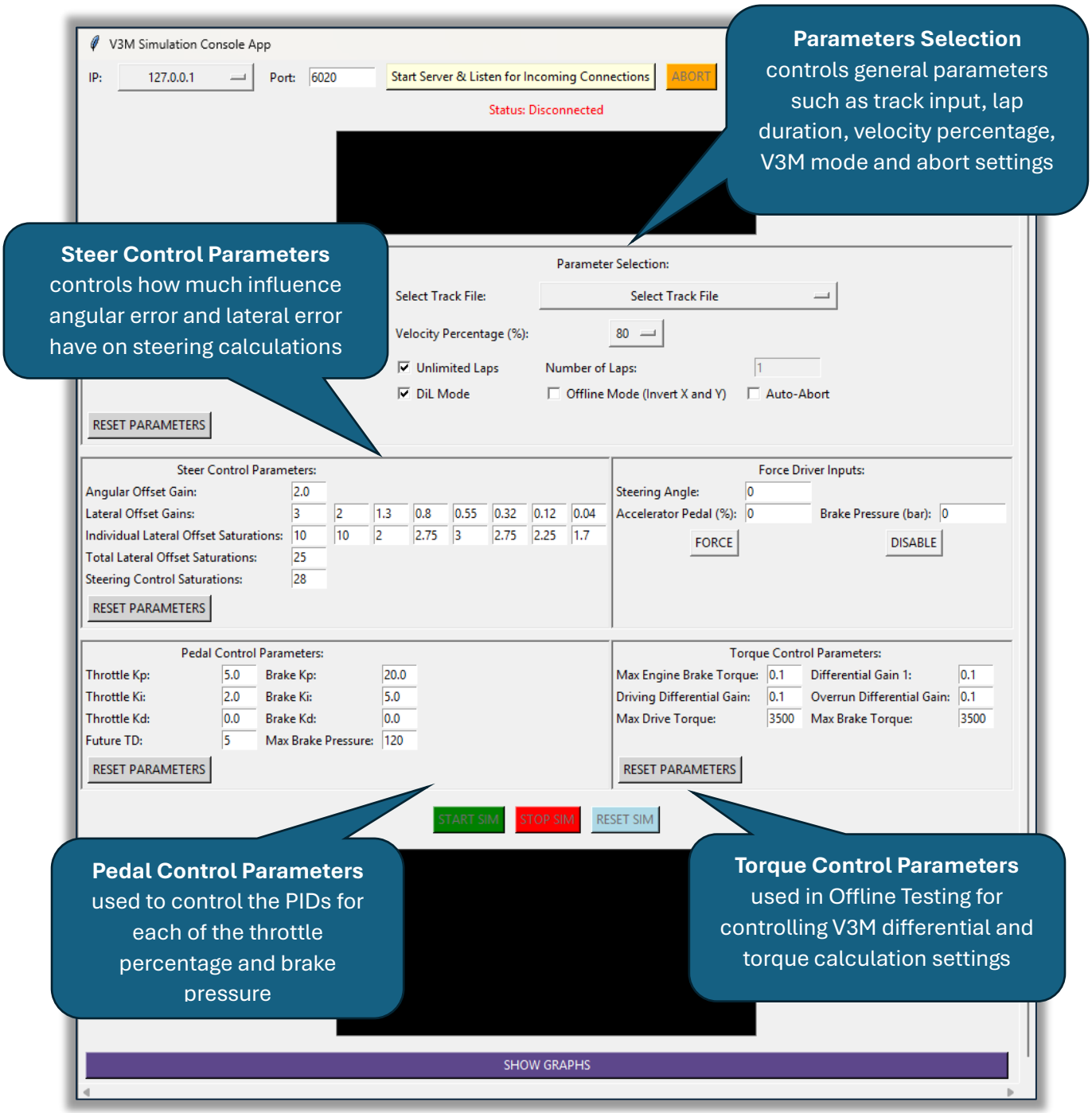


Figure 7: Parameter Window Overview

8. Running the Simulation

8.1. Start Button

- Once a connection is active and the simulator vehicle is initialised, press the **"START SIM"** button to begin the simulation.

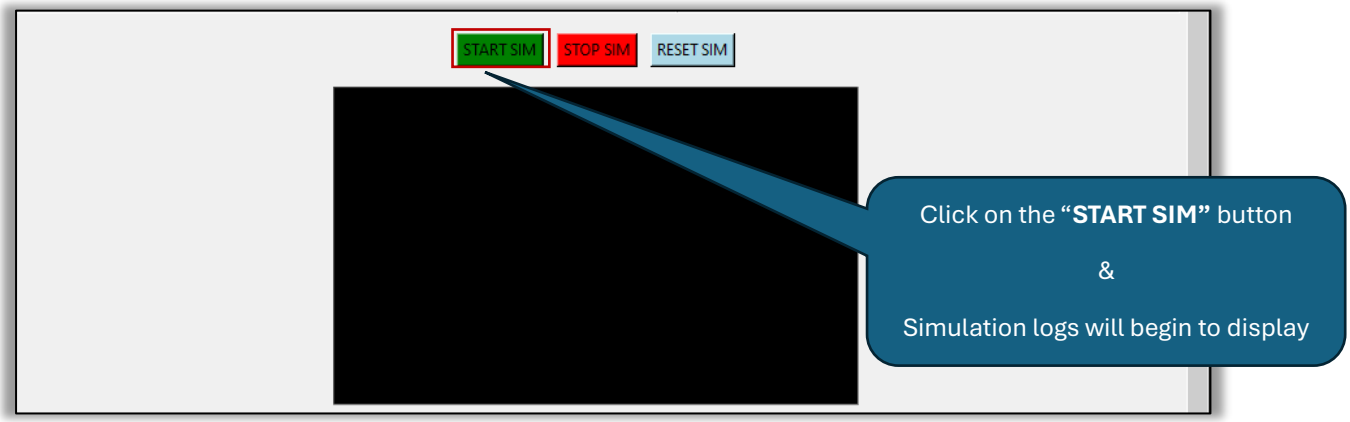


Figure 8: Running the Simulation - Start Button

- The V3M initiates "Start Up" Mode, wherein the throttle percentage increases at a constant rate, until the target velocity profile is matched
- Simulation data will begin flowing in real time and be displayed in the console output.
- Vehicle state, control signals, and metrics will be recorded for plotting.
- This button is also made accessible again after the stop button is pressed, or when a new connection is established

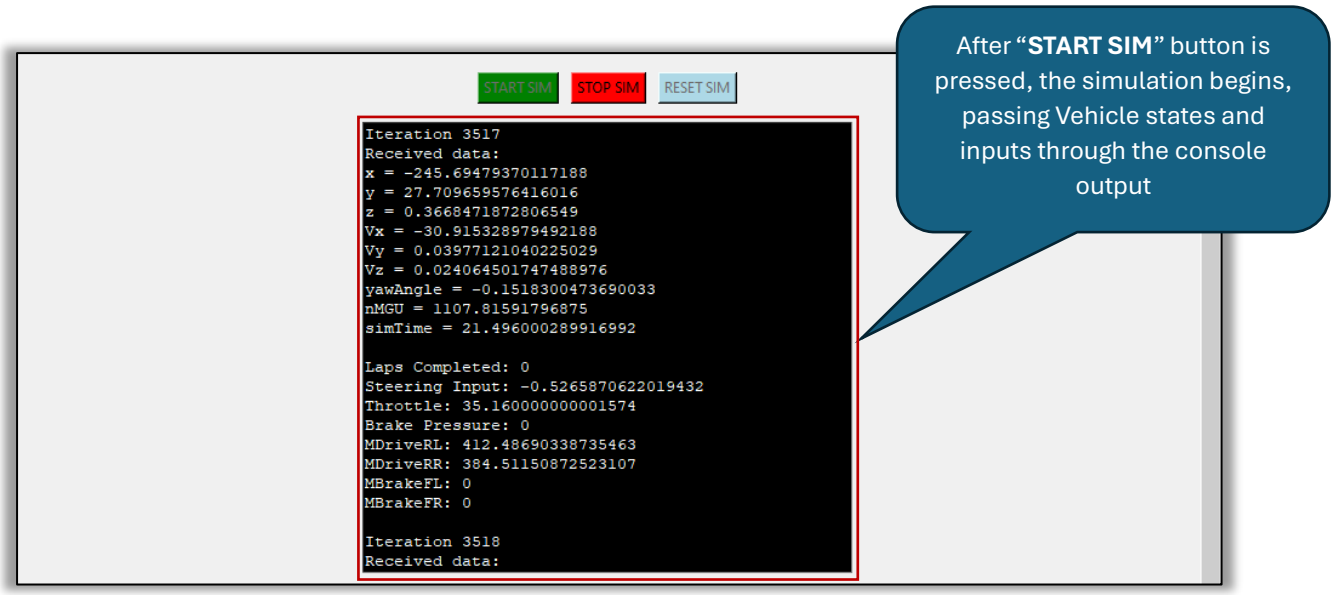


Figure 9: Console outputs Simulation Data

9. Simulation Control Buttons

9.1. Stop Button

- Enabled only during a simulation run. (By default, it is disabled when the simulation is not running)

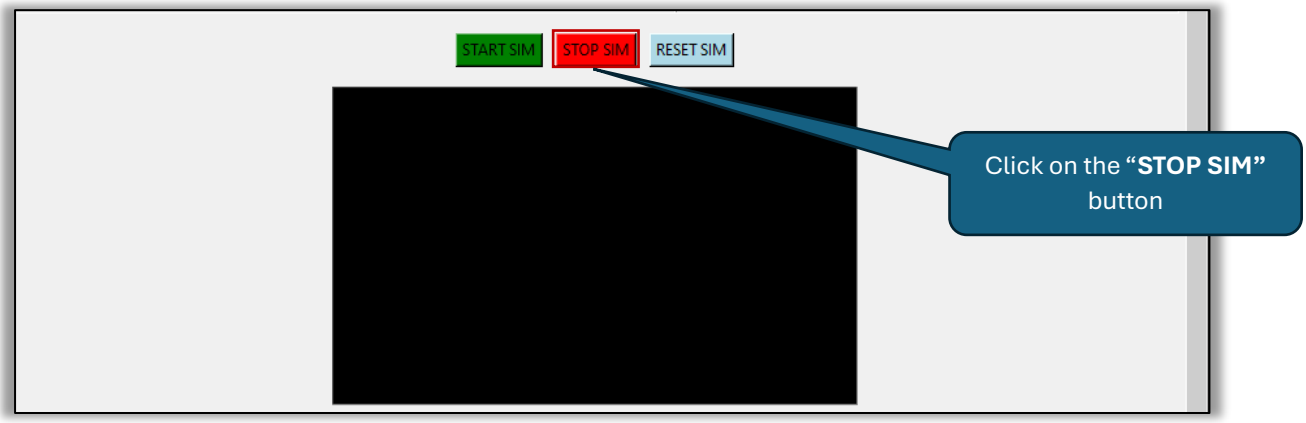


Figure 10: Simulation Controls - Stop Button

- "**STOP SIM**" Button Functionality - Sends maximum brake pressure to the DiL simulator.
- The "**START SIM**" button will be enabled for starting the V3M again (if selected, Start Up Mode will initialise again)

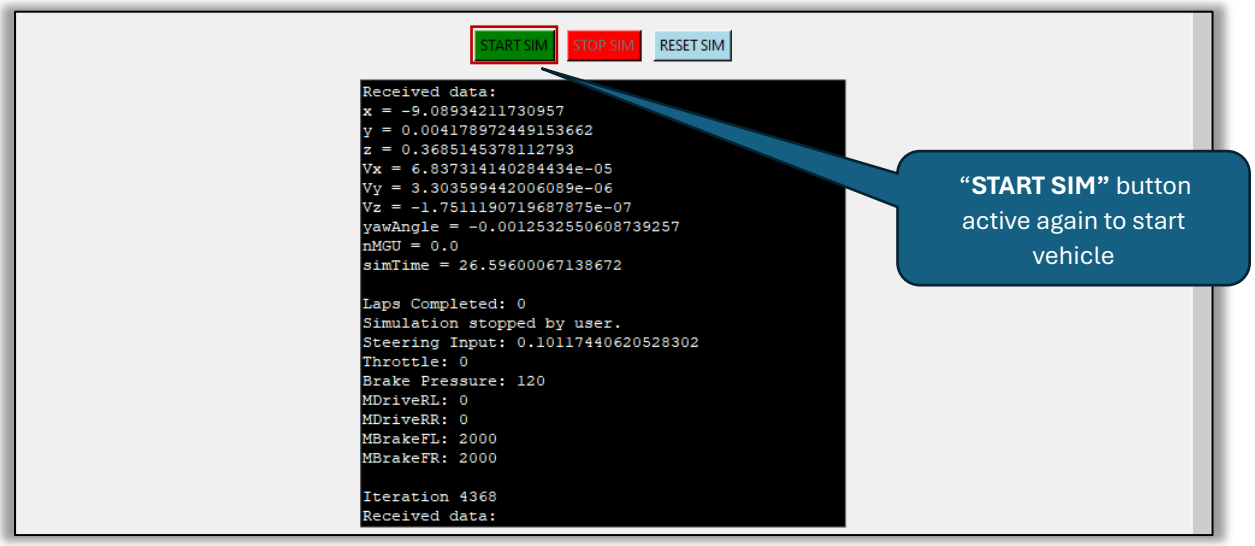


Figure 11: Restarting the V3M

9.2. Reset Button

- Enabled only during a simulation run and whilst the simulator vehicle has stopped after pressing "**STOP SIM**".
- "**RESET SIM**" Functionality - Closes the connection safely, saves all plots and resets V3M Simulation Console to original state.

Figure 12: Simulation Controls - Reset Button

9.3. Abort Button

- "**ABORT**" Button Functionality - Immediately disconnects the application from the simulator.
- Causes the simulator to come to a full stop, saves all plots and resets V3M Simulation Console to original state.
- Intended for emergency use or quick termination.

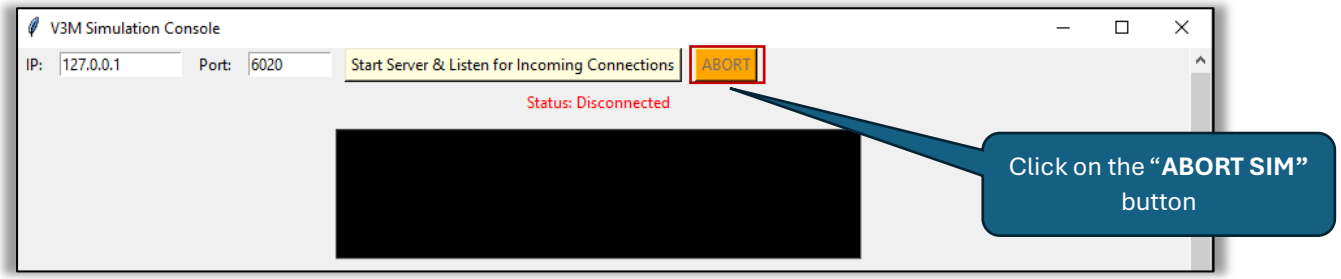


Figure 13: Simulation Controls - Abort Button

10. Plotting and Visualisation

Once the simulation completes or is stopped, the application automatically processes and plots key data such as:

- Vehicle trajectory and velocity profile
- Steering and pedal control signals
- Yaw rate, reference distance, orientation and angular/lateral/velocity errors

10.1. Viewing the Graphs

- Generated plots will appear in a separate window using matplotlib (*Matplotlib documentation — Matplotlib 3.10.5 documentation*, 2025)
- You can zoom, pan, or save them manually using the matplotlib toolbar.
- For a more detailed overview of the outputted graphs please see Live Data Logging and Graph Exports (under Section 13.3)

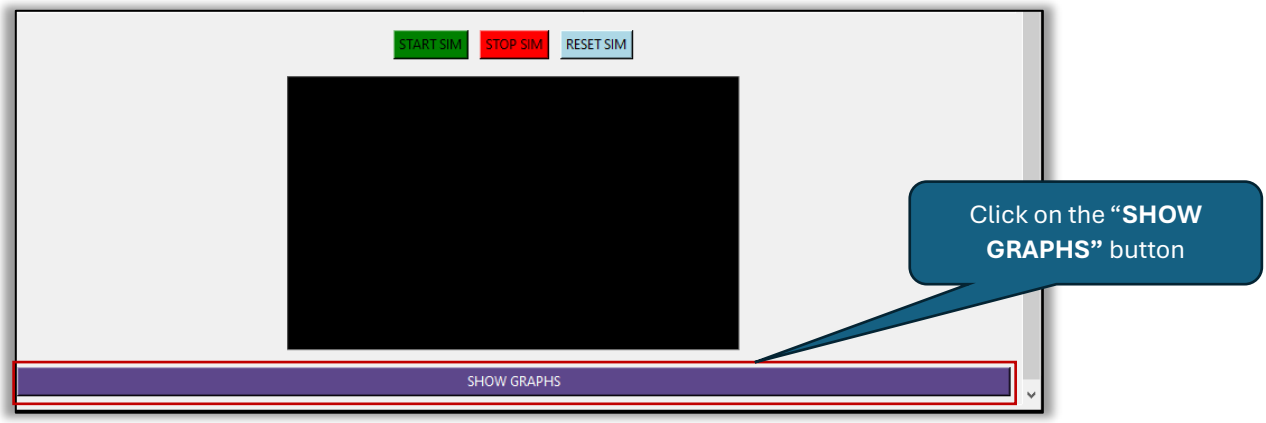


Figure 14: Show Graphs Button

10.2. Saving and Displaying Results

- After the simulation ends, the application automatically saves all the graphs by default:
  - Graph images (.png files)
- Once the "**SHOW GRAPHS**" button is clicked a separate "**Display Simulation Graphs**" window pops up for viewing various graphs.
- When "**Display Simulation Graphs**" window appears, hovering the mouse over the images shows a zoomed preview of the same graphs.
- Double clicking the zoomed in preview of the same graph pops up for viewing the individual graph in a full screen window
- Alternatively, the "**Select All**" button will select all graphs and pressing "**Display Selected Graphs**" will display all graphs. In addition, checking the adjacent box and pressing "**Display Selected Graphs**" will display the individual graphs. The graphs will display at the bottom at of the window.

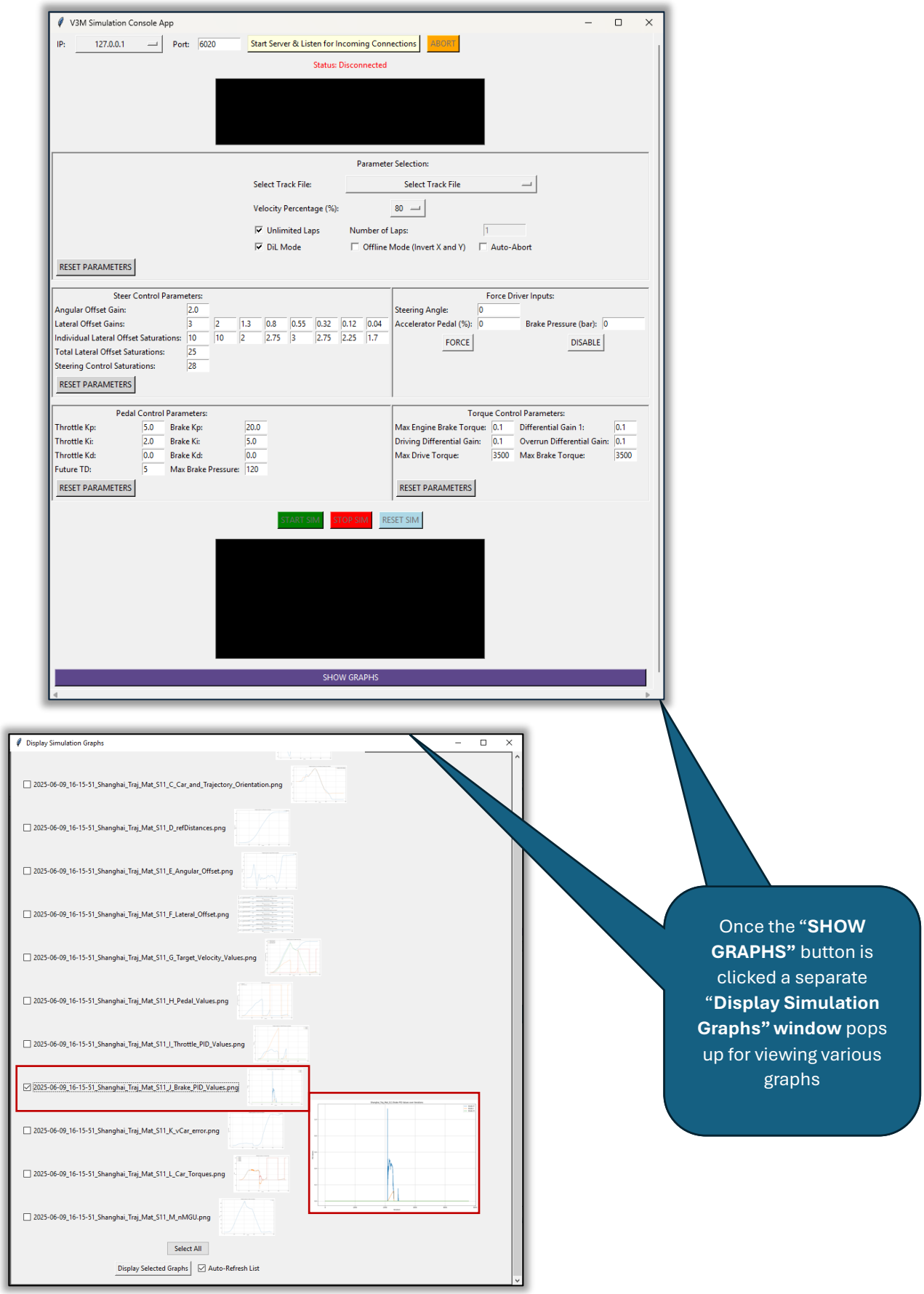


Figure 15: Display Simulation Graphs Window

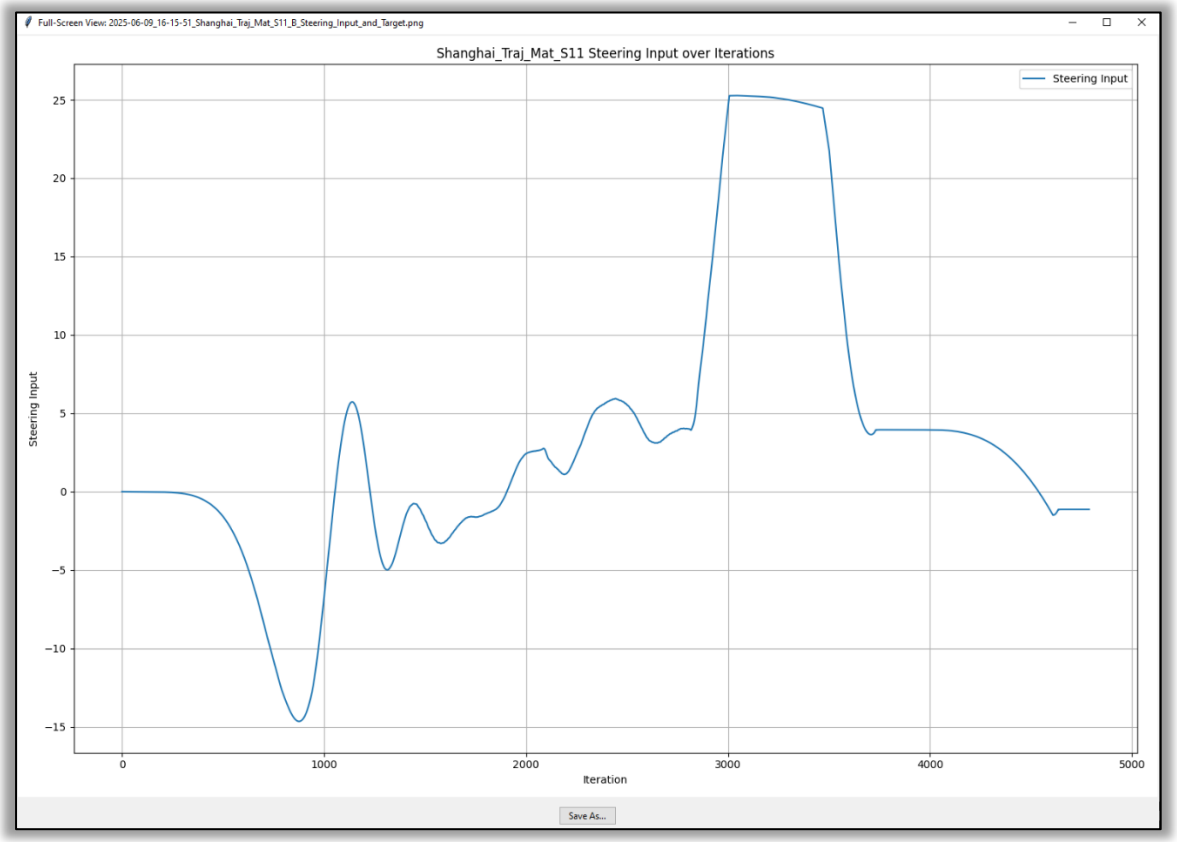


Figure 16: Full Screen View for Individual Graph

11. Simulation Modes: DiL (Real-time) and Offline Modes

The application supports two primary simulation modes. These modes influence how trajectory and control data are generated and interpreted.

11.1. DiL Mode (Driver-in-the-Loop)

- Used only when interfacing with the physical simulator hardware (DiL Simulator).
- Vehicle response must be smooth and in real-time.
- Application processes input from hardware and adapts the driving line accordingly.

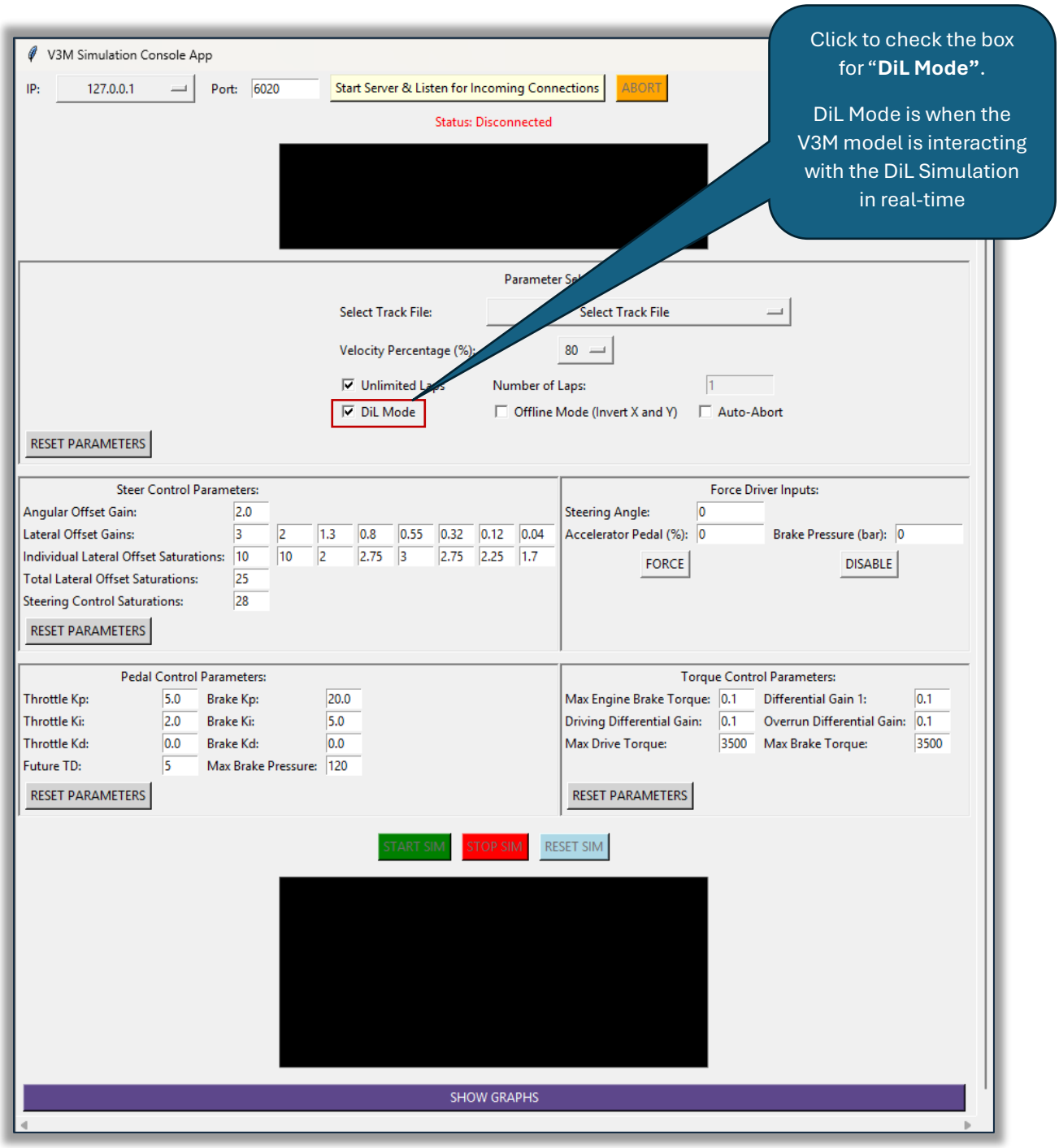


Figure 17: Check Box to run on DiL Mode (Drive-in-the-Loop)

11.2. Offline Mode

- Used in offline, software-only environments for developmental work independent of the DiL Simulator. (*Discrete-Event Simulation - MATLAB & Simulink*, 2025)
- Allows running repeatable tests on a local machine.
- Simulations may run faster than real-time depending on setup.
- Minor differences include:
    - Inverting x and y coordinates
    - Transforms input data starting x and y coordinates to (0, 0)
    - Scales steering input from metres to millimetres

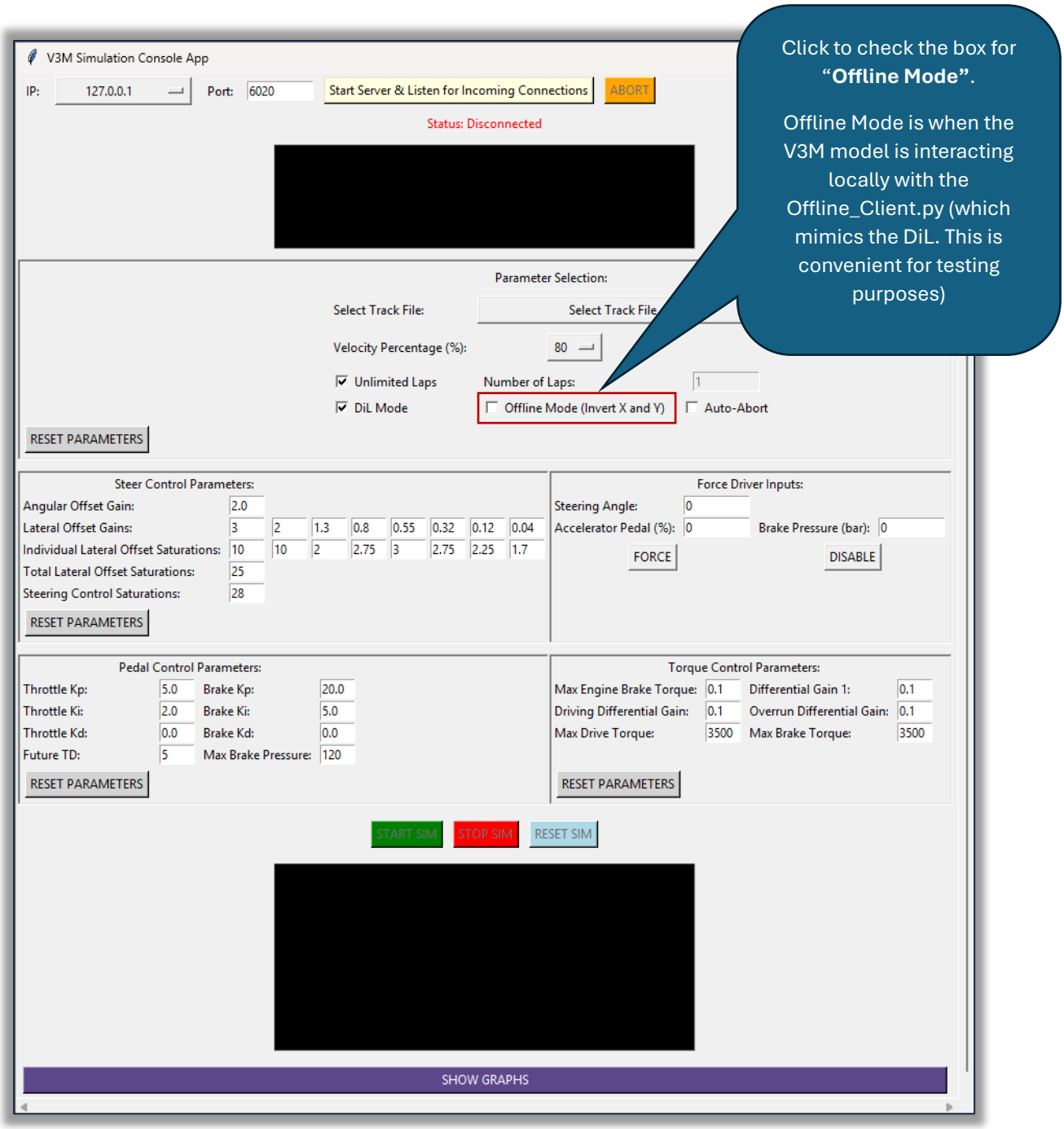


Figure 18: Check Box to run on Offline Mode (Locally interacting with an Emulator)

12. Using the V3M Application: Step-by-Step Instructions

Follow the sequence below to operate the V3M simulation model in coordination with the DiL simulator:

Step 1: Launch the Application

- Double-click the V3M_ExtDriver.exe file to open the application.
- The interface will appear, allowing you to configure settings and prepare the model.

Step 2: Configure Network Settings

- Enter the simulator IP address (x.x.x.x) and port (xxxx) into the provided fields.

Step 3: Start the Server

- Click the "**Start Server & Listen for Incoming Connections**." button
- The application will wait for the DiL simulator to connect.
- Once connected, a confirmation message will appear in the console output.

Step 4: Load Input File

- Load the required track when prompted or through the "**Select Tracks**" parameter.
    - Track file: Contains path matrix data as a .csv file (e.g., reference distance, x, y coordinates, vehicle velocity)
- The V3M will use this information to generate a reference driving line for the selected track.

Step 5: Review and Adjust Parameters

- Enter or adjust driver model parameters (e.g., mass, max torque, control gains) as needed.
- If unsure, use default values provided by the application.

Step 6: Start the Simulation

- Once the simulator is connected and parameters are set, click "**START SIM**"
- The application will begin calculating control commands in real time:
    - Steering angle (xSteerRack): Computed using a look-ahead path preview system based on the reference line (Casanova, 2000)
    - Accelerator Pedal Percentage (rAccel) and Brake Pressure (pBrakeFIA): Uses PID controllers and velocity error against the target velocity profile (*The PID Controller & Theory Explained*, 2025)

Step 7: Monitor Simulation Progress

- The DiL simulator receives control signals and returns updated vehicle states.
- These are displayed in the console and used to compute the next control inputs continuously.

Step 8: Control the Simulation as Needed

- ABORT: Immediately stops communication and simulation

- STOP SIM: Gracefully halts the simulation but keeps the connection and simulation running
- RESET SIM: After pressing "**STOP SIM**", closes the connection and returns V3M GUI to default state

Step 9: View and Save Results

- Once the simulation stops, results will be processed and plotted automatically.

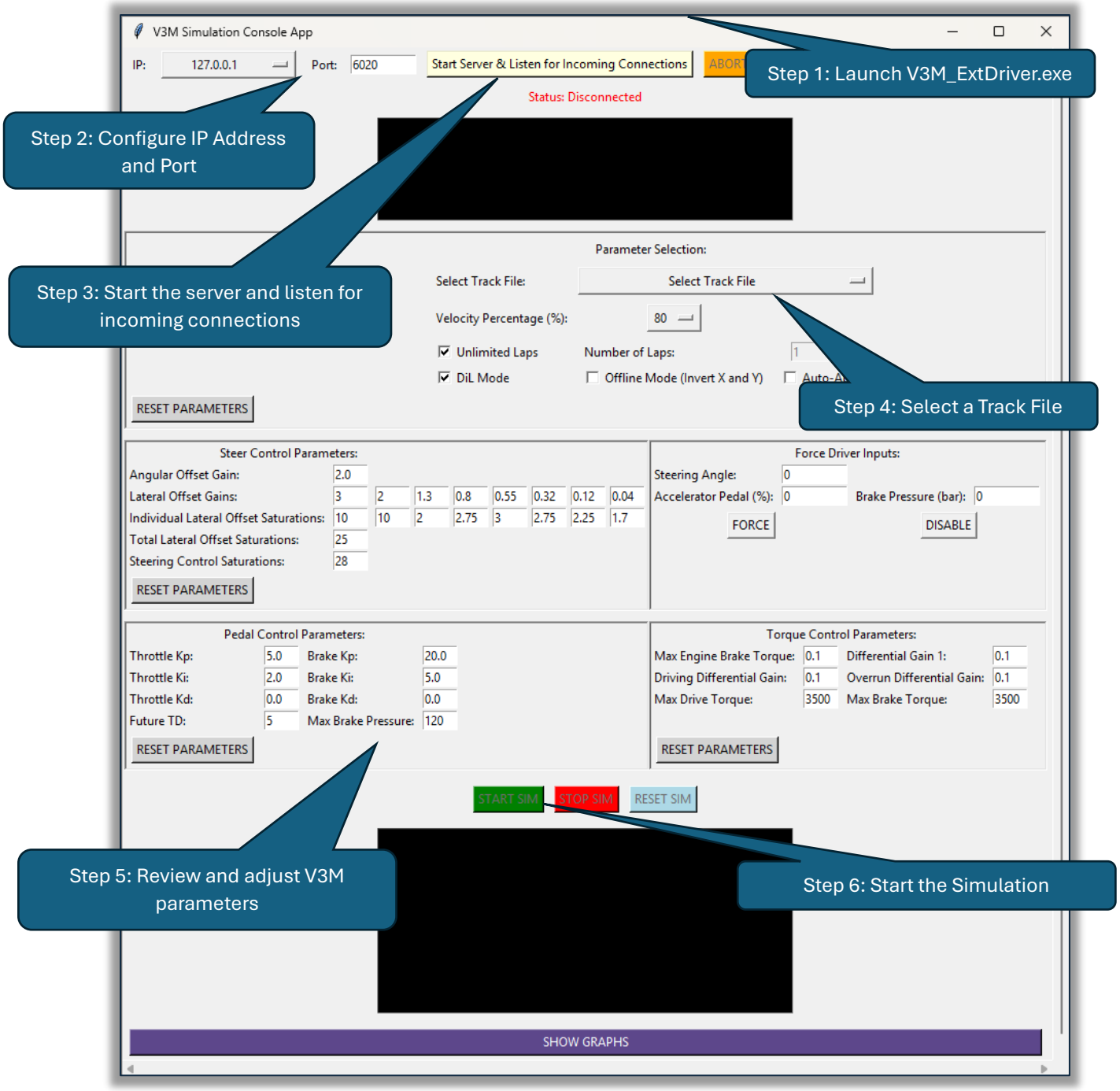


Figure 19: Step-by-step Guide – Parameter Screen

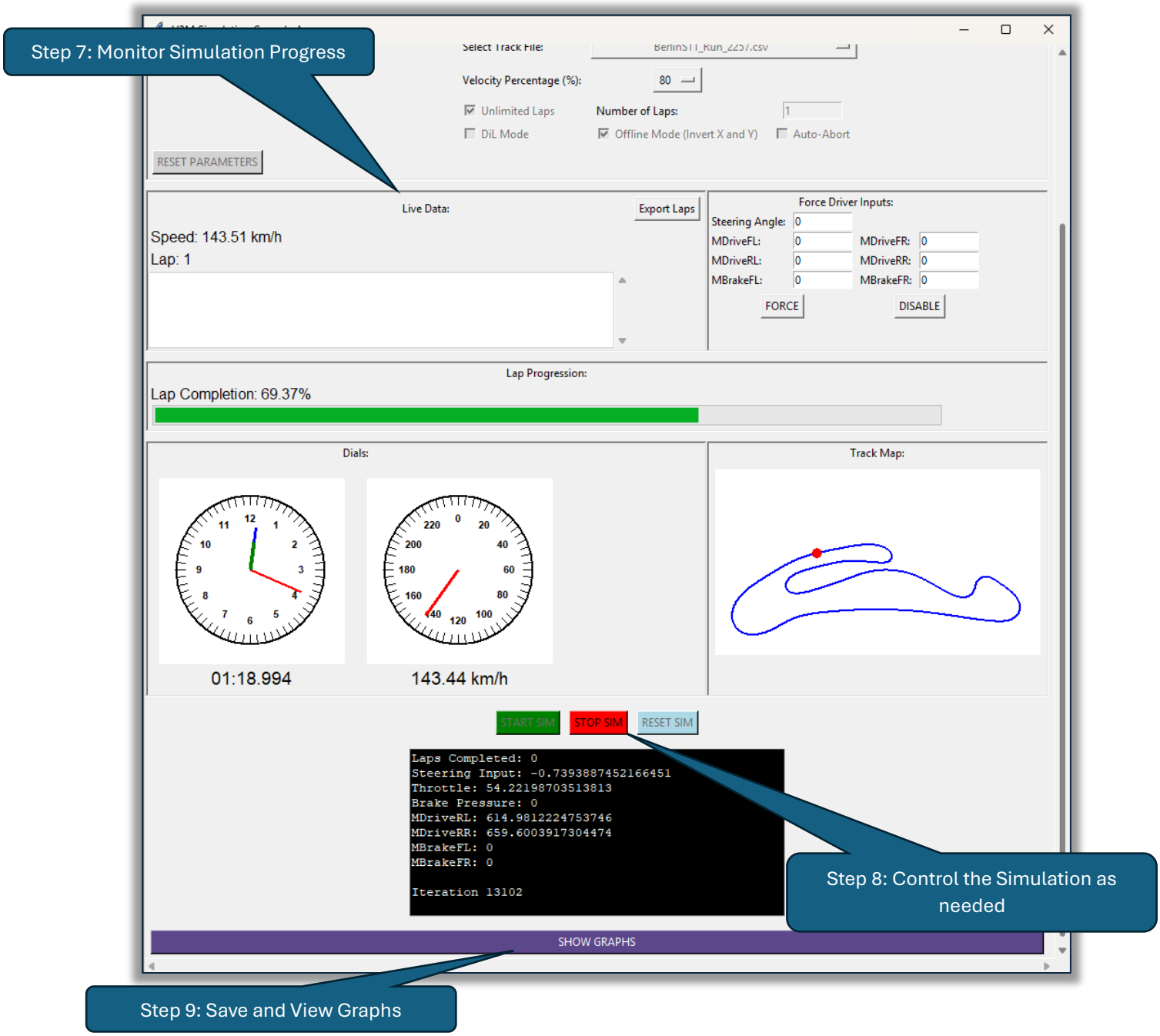


Figure 20: Step-by-step Guide - Live Data Screen

13. Live Data Logging and Graph Exports

13.1 Live Data Display

The live data on display during simulation runs includes:

1. The Live Data Window:
    - Current Vehicle Speed (in kph)
    - Current Lap Counter
    - Previous Lap Times
        - Previous Lap Times here may be exported to a separate file using the button "**EXPORT LAPS**"
2. The Lap Progression Window:
    - A Lap Completion progress bar
3. The Dial Window:
    - Current Vehicle Speed in dial format in kph
    - Current Lap Time in dial format
4. The Track Map Window:
    - Live Track Map Display with updating Vehicle Icon

13.2 Force Driver Inputs

- Force Driver Inputs remain active during the simulation run
- "**FORCE**" – Overrides all driver model inputs and sends custom inputs from the designated fields
- "**DISABLE**" – Reactivates driver model

13.3 Live Adjustable Velocity Percentage

- The Velocity Percentage Adjuster remains activated for a simulation run to allow modification velocity speed throughout a simulation run
- NOTE: Changing velocity during corners may result in the vehicle from spinning out on the track

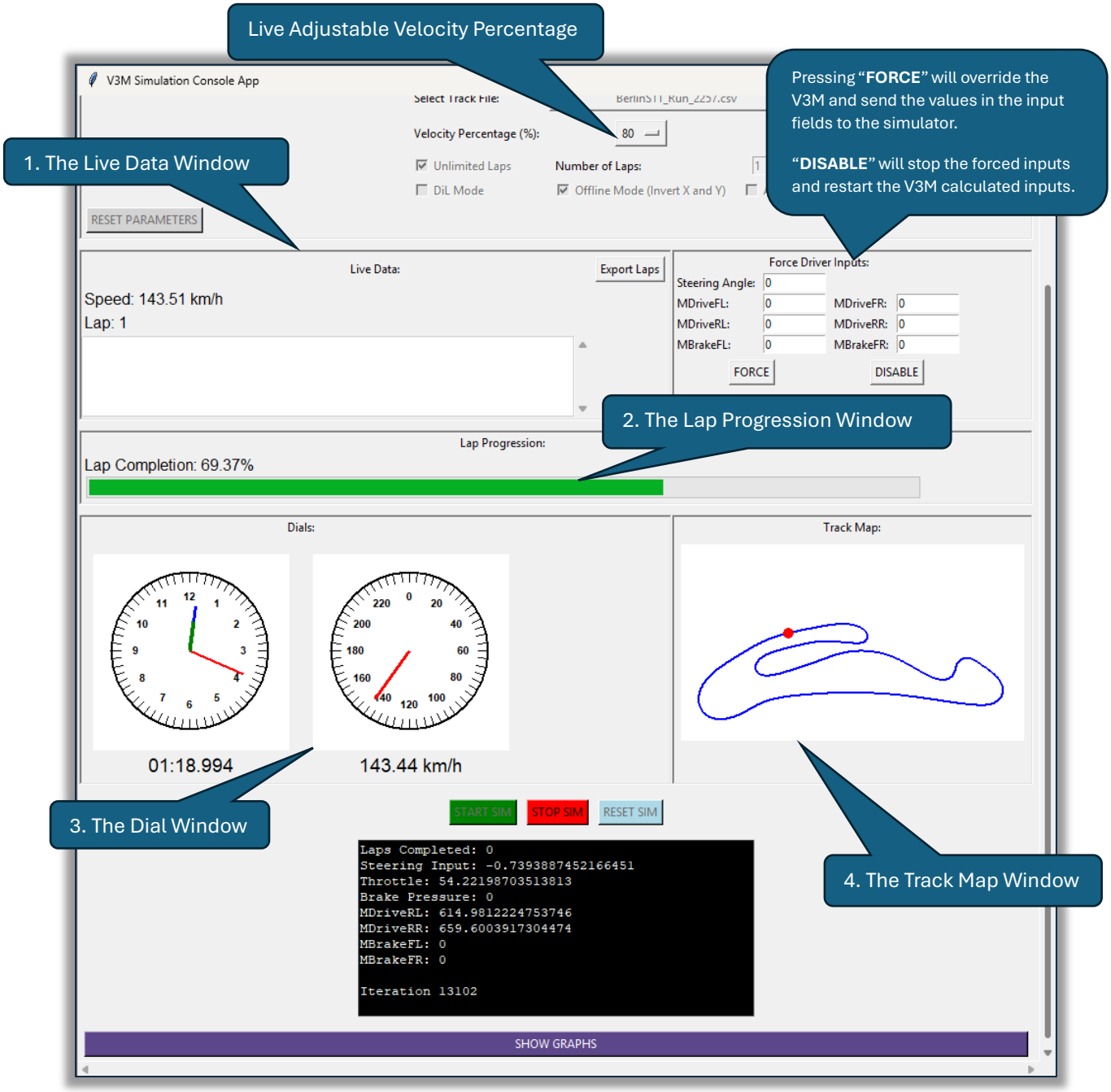


Figure 21: Live Data Display During Simulation Run

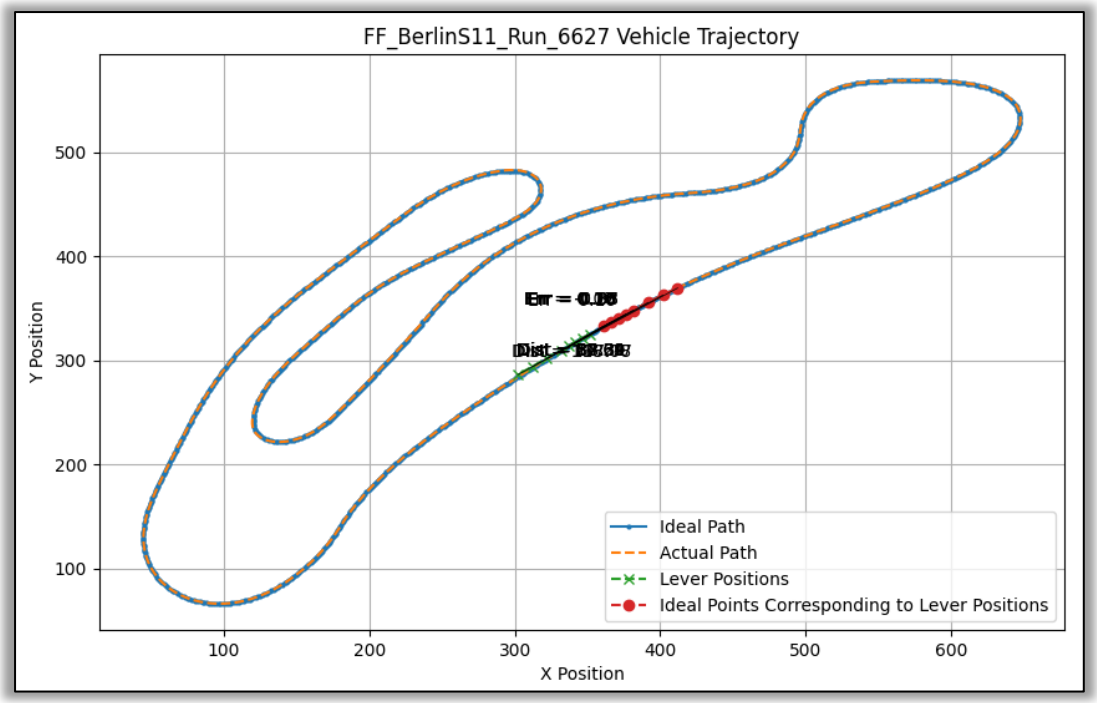
13.3 Graph Export

Following a simulation run after the "**ABORT**" or "**RESET SIM**" button is pressed, several graphs will be saved to the "Simulation Graphs" folder. These graphs are available to view using the "**SHOW GRAPHS"** button (as seen in Section 10.2 Saving and Displaying Results). The V3M will save the following graphs:

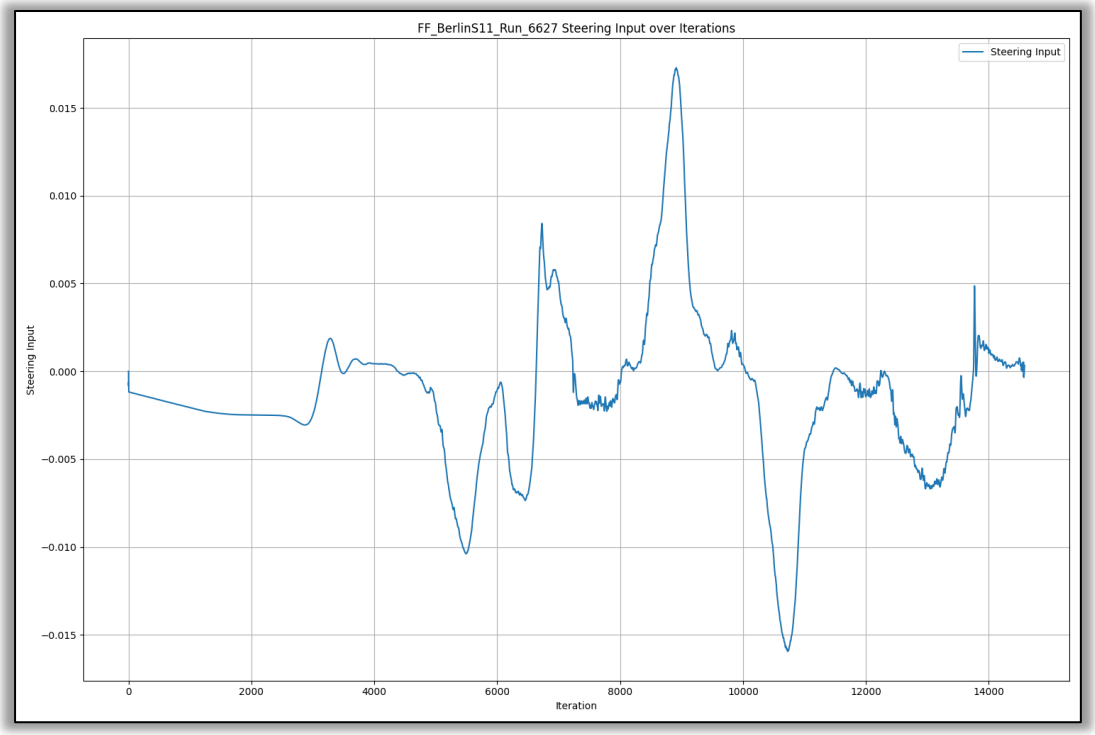
1. Vehicle Trajectory
2. Calculated Steering Input
3. Vehicle Orientation vs Track Orientation
4. Reference Distance Trace in Metres
5. Angular Offset of the Vehicle
6. Lateral Offsets of the Vehicle
7. Target Velocity vs Current Velocity with Throttle and Brake Input Overlap
8. Throttle Percentages and Brake Pressure Inputs
9. Throttle PID Values
10. Brake PID Values
11. Total Velocity Error
12. Car Torque Inputs
13. nMGU (Motor Speed) Values

1. Vehicle Trajectory

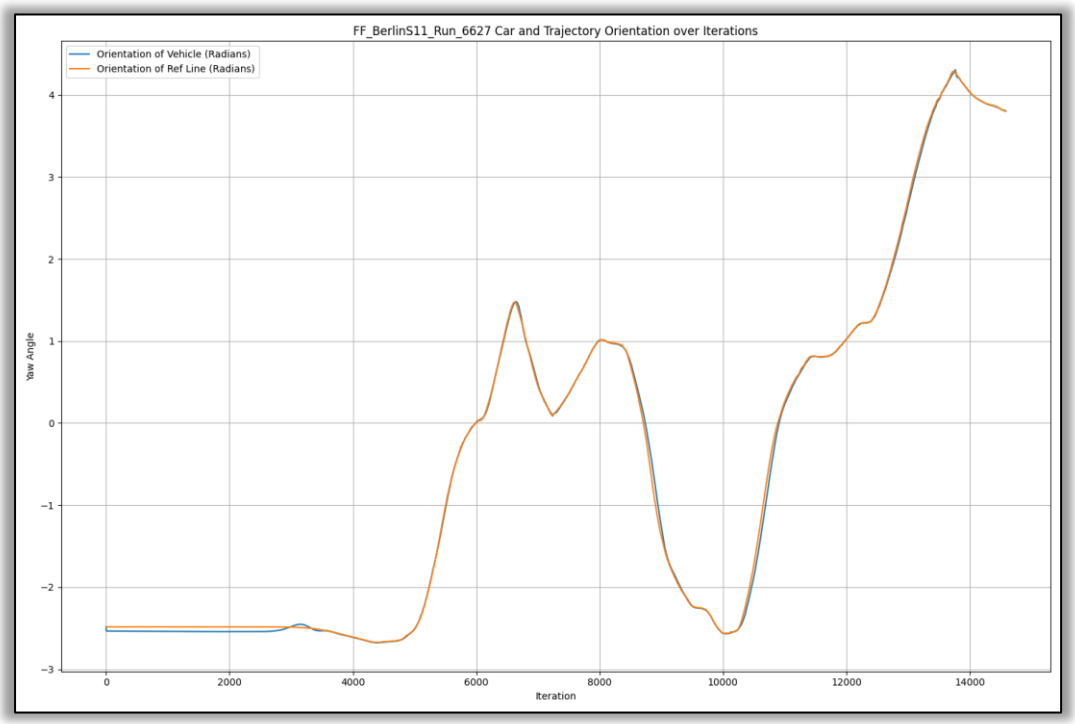
- V3M trace against inputted ideal path
- Also includes Vehicle Lever Positions against Ideal Points on the Lap Trajectory and lateral error values



2. **Calculated Steering Input (DiL mode in m, Offline mode in mm)**
   - The Calculated Steering Rack value submitted to the DiL Simulator every iteration (~0.002 seconds) in metres
   - Offline mode takes input in millimetres

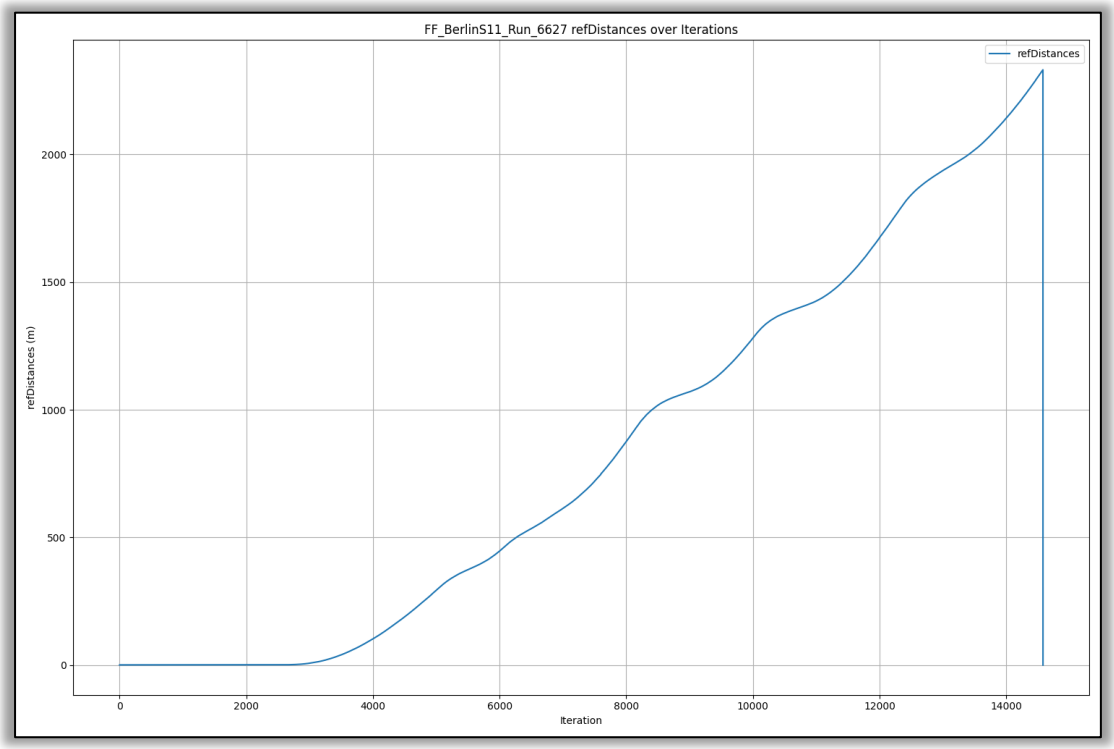


3. **Vehicle Orientation vs Track Orientation**
   - Offers a comparison in radians of the orientation of the vehicle against the reference line's orientation



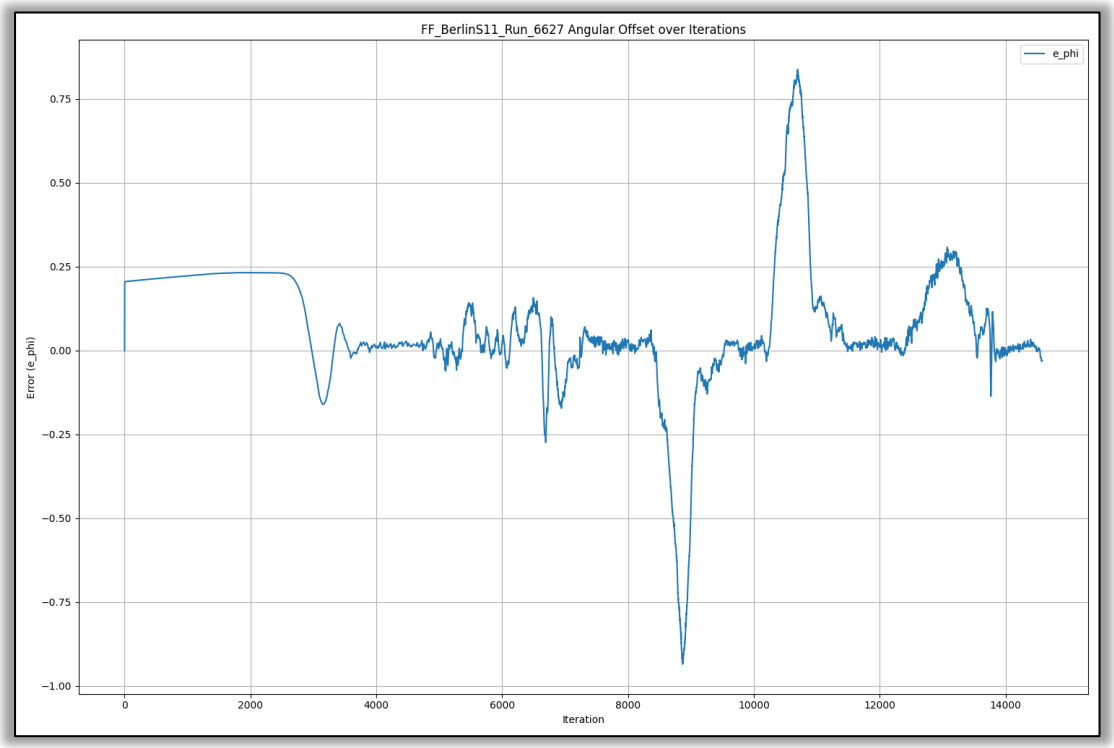
4. Reference Distance Trace

- Offers a reference distance trace for each lap of the simulation

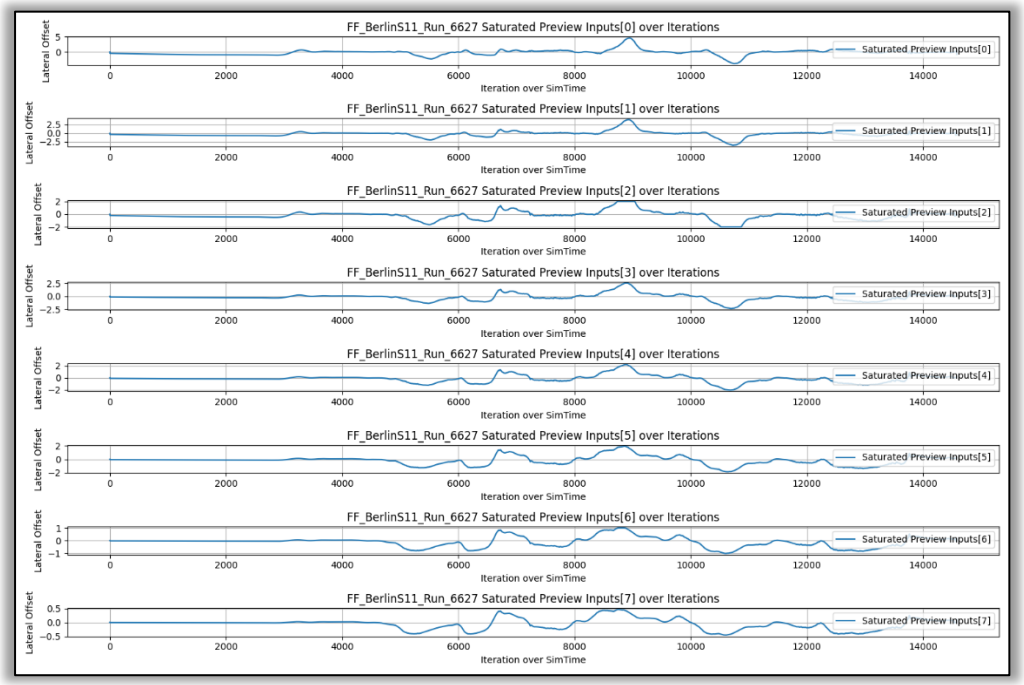


5. Angular Offset of the Vehicle

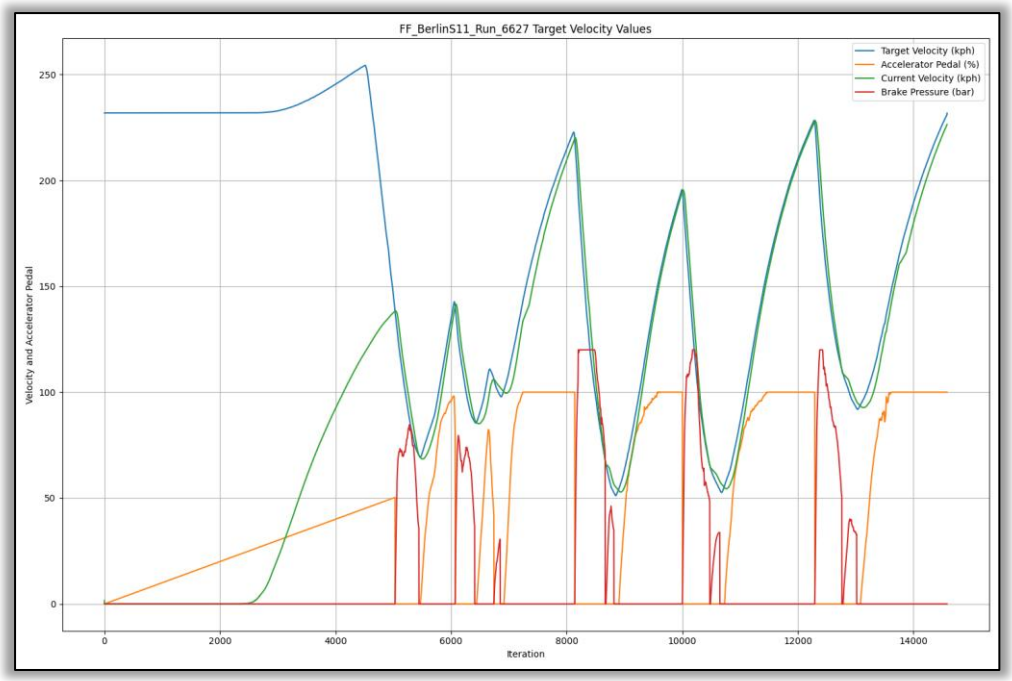
- In depth look at the angular error difference between the Vehicle and Trajectory in Radians



6. **Lateral Offsets of each Lever Position from the Vehicle**
   - Individual lateral errors in metres at each position on the optical lever beginning with “Steering Preview Inputs [0]” being the lateral error of the vehicle itself
   - The Optical Lever refers to the distance ahead of the vehicle that the V3M uses to respond to lateral deviations, and this distance scales with the vehicle's speed (Casanova, 2000)

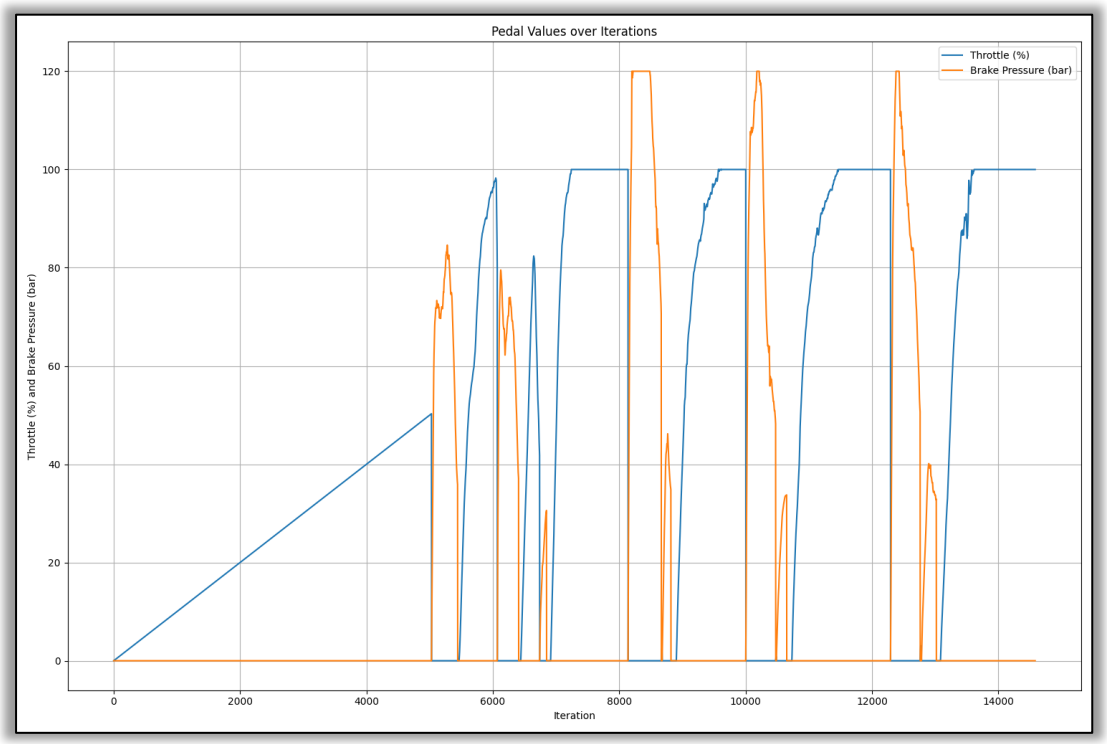


7. **Target Velocity vs Current Velocity with Throttle and Brake Input Overlap**
   - Offers a comprehensive overview of the speed control within the V3M
   - Adds a comparative view of the vehicle speed against inputted speed of the target velocity profile in kph
   - On top of this comparison, the graph includes the inputs for throttle and brake inputs against the vehicle’s actual and expected velocity



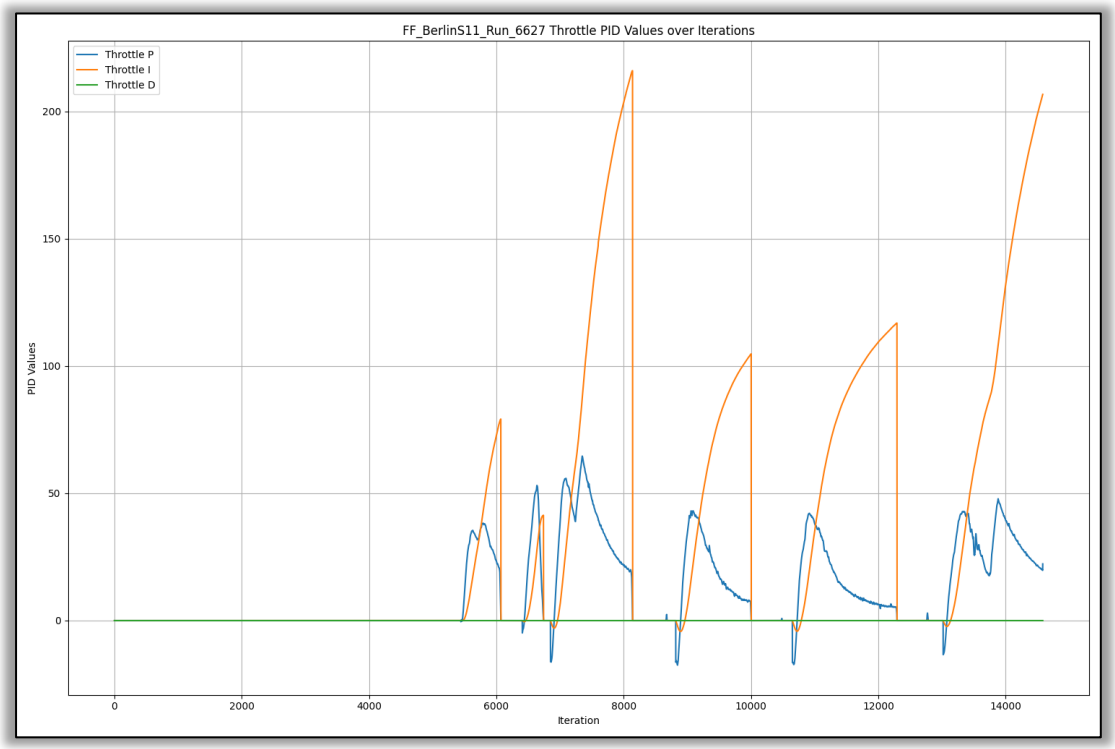
8. Throttle Percentage and Brake Pressure Input

- Dedicated graph looking into accelerator pedal percentage vs brake pressure (bar)



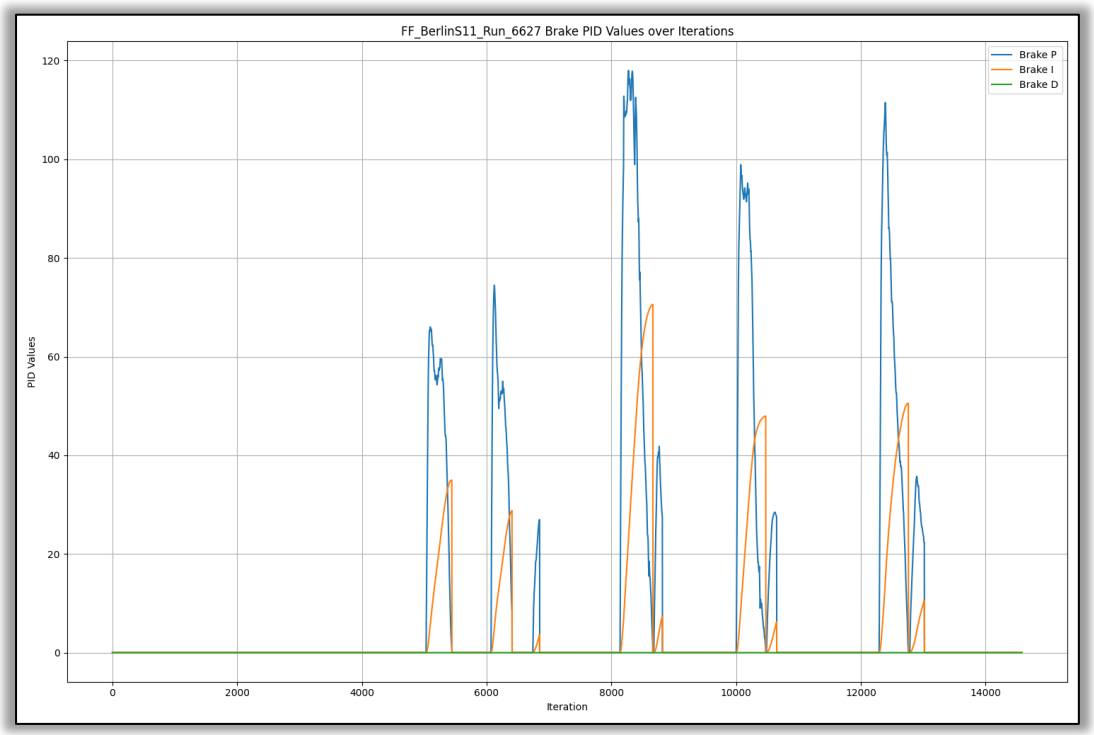
9. Throttle PID Values (using velocity error)

- Each P, I and D value which, summed up, are responsible for throttle percentage calculation:
  - P (Proportional): reacts to current velocity error
  - I (Integral): corrects past velocity error
  - D (Derivative): predicts future velocity error to smooth response



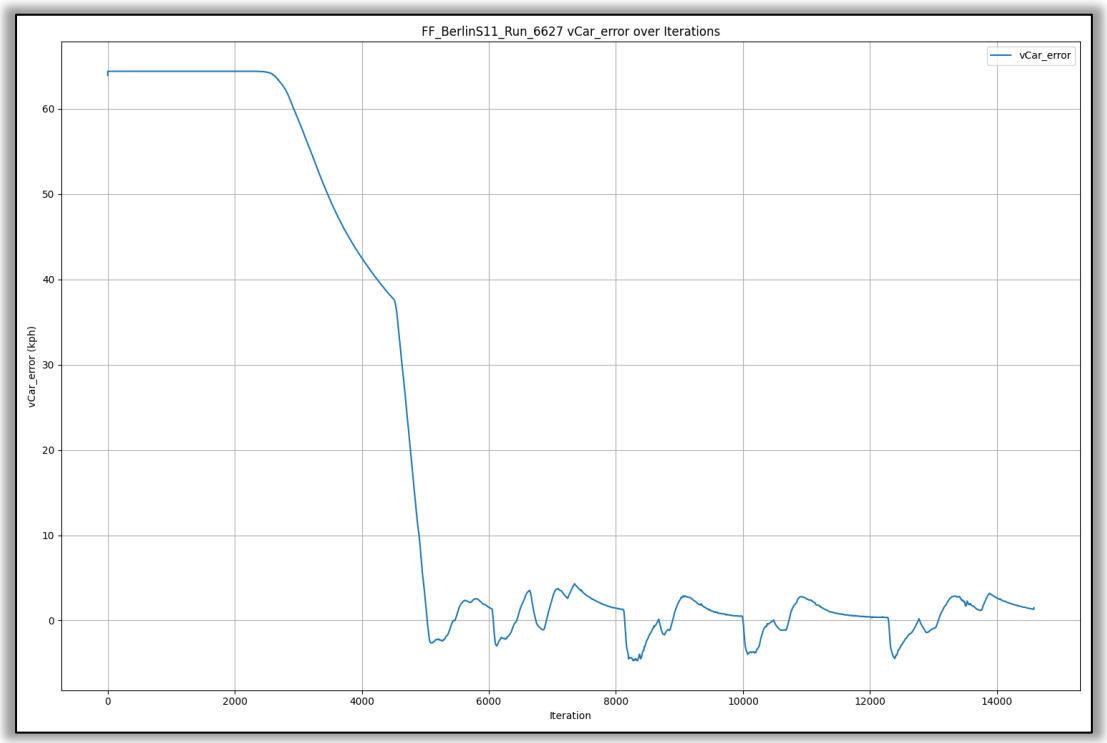
10. Brake PID Values (using velocity error)

- Each P, I and D value which, summed up, are responsible for brake pressure calculation



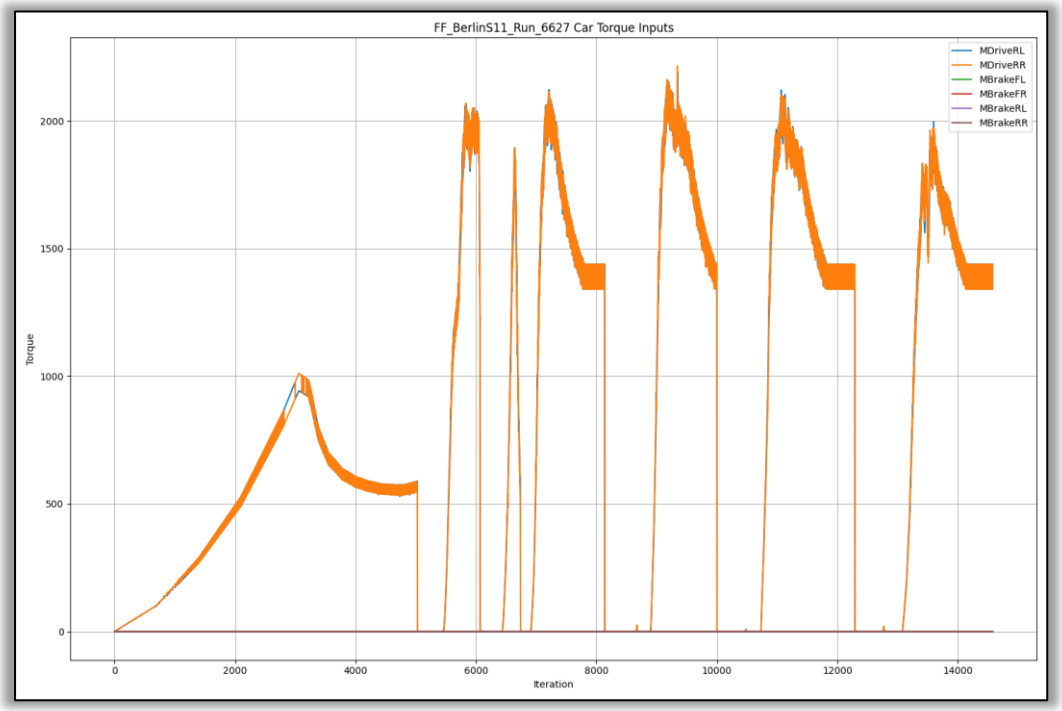
11. Total Velocity Error

- In depth look at the velocity error difference between the Vehicle and Target Velocity Profile in kph



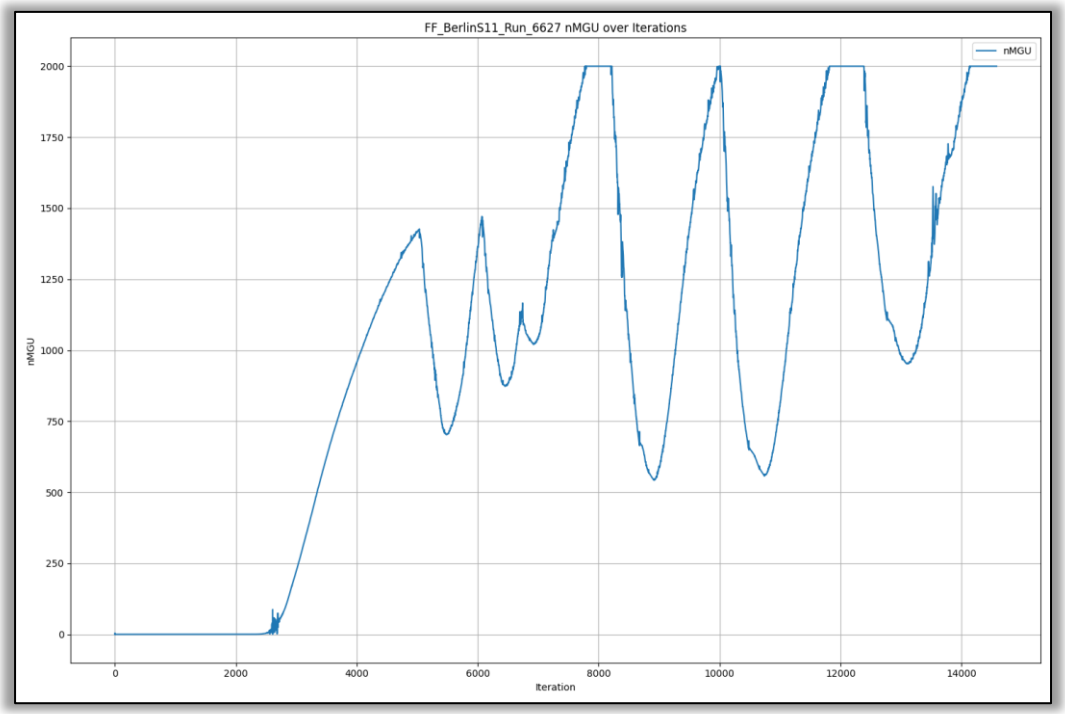
12. Driveshaft/Brake Torques

- As calculated from the V3M's internal Pedal Map
- ONLY APPLICABLE in Offline Mode



13. nMGU Values (Motor Speed Values)

- Motor speed values required to generate the appropriate driveshaft torque with the V3M's internal pedal map
- ONLY APPLICABLE in Offline Mode



14. Experiments, Performance Evaluation and Metric

The V3M application is engineered to autonomously operate a Driver-in-the-Loop (DIL) simulator, enabling efficient testing and development of vehicle control strategies without the need for a human driver. This section outlines the performance metrics used to evaluate the model, along with enhanced simulation capabilities designed for comprehensive testing.

Objectives of Experiments and Evaluation

The V3M is assessed with the following objectives in mind:

1. **V3M Autonomous Driving Simulator - Unattended Operation:** Enable fully autonomous driving within the simulator to save operator time and resources.
2. **Developmental Software Changes - Development Support:** Provide a consistent, repeatable driver input source to validate software updates and tuning changes in the V3M to optimise Vehicle Control Module (VCM) testing.
3. **Velocity Performance Targeting:** Achieve high-fidelity driving behaviour, with velocity targets.
4. **Flexible Performance Simulation Control:** Allow testers to run the model over unlimited laps and inject custom driver inputs into the simulation pipeline for hybrid or override testing.

Metric	Description	Target
Lap Time Consistency	Deviation in lap time across consecutive runs. Measured over unlimited laps if required.	Variation in lap time ≤ 0.75% after out lap (1 lap)
Developmental Software Changes - Development Support	Validating software updates and tuning changes in the V3M with different parameter setups.	Demonstrate V3M flexibility through controlled stability, velocity error reduction and differential control
Velocity Performance Targeting	Achieving 100% velocity in high performance environments	Current velocity reaches target velocity; Zero crash outs
Flexible Performance Simulation Control: Override Compliance	Ability of the V3M to accept and blend forced driver inputs injected via simulation.	Deviation from target reference line

Table 2: Performance Metrics & Evaluation Criteria

14.1. Experiment 1: V3M Autonomous Driving Simulator - (Lap-Time Consistency)

Goal: To validate that V3M to enable full autonomous driving within the simulator to save operator time and resources, while maintaining lap-time consistency.

14.1.1 Experiment Scenario/Design:

- Track: Same track (Tokyo) used across to test autonomous driving and consistency of laps
- Laps: 1, 5 and 20 continuous laps (configured with set parameters), no restarts unless safety triggers.
- Mode: Run in Dev Mode and DiL mode.

14.1.2 Experiment Parameters:

Parameter	Description	Default / Range	Notes
Track Selection	Same track used across to test autonomous driving and consistency of laps	Tokyo with 100% velocity	Keep constant across runs
Velocity Profile	Target speed vehicle must achieve per distance marker	Default	By default set by system
Steer Control Parameters	Angular Offsets, Lateral Offsets, Individual Lateral Offset Saturations and Steering Control Saturations	Default	By default set by system
Pedal Control Parameters	Throttle Kp, Throttle Ki, Throttle Kd, Future Target Distance (TD)	Default	By default set by system
Torque Control Parameters	Max Engine Brake Torque, Driving Differential Gain, Max Drive Torque	Default	By default set by system
Forced Driver Inputs	Steering Angle, Accelerator Pedal, Brake Pressure, MDriveFL, MDriveFR, MDriveRL, MDriveRR, MBrakeFL, MBrakeFR	Default	By default set by system
Autonomy Mode	Race Car Control setting	Full V3M autonomy	No human intervention
Lap Count	**Number of consecutive laps**	**Min: 1, 5 and Max: ≥ 20 laps**	**First lap is out lap**
Sim: Environment	Other cars or obstacles	Off	Avoid external variability
Logging Rate	Sampling frequency	≥ 100 Hz states	Ensure disk throughput is sufficient
Incident Auto-Reset	Automatic recovery after incident	Enabled	Resume next lap without manual input
Acceptance Thresholds	Targets to judge success	Variation in lap time ≤ 0.75% (across all laps)	Used for evaluation; not a runtime input

Table 3: Parameters used for Experiment 1

14.1.3 Experiment Procedure/Methodology:

1. Load reference track.

2. Start Server & Listen; Start Simulation (V3M in full autonomous mode).
3. Independent Variables: Lap count 1, 5 and 20 laps
4. Enable automatic session logging and telemetry export.
5. Allow continuous operation; stop only on completion or safety abort.

14.1.4 Experiment Results:

Lap Count	Lap Time with 1 Lap	Lap Time with 5 Laps	Lap Time with 20 Laps
1	01:39.1	01:39.1	01:39.1
2		01:32.1	01:32.1
3		01:32.1	01:32.1
4		01:32.1	01:32.1
5		01:32.1	01:32.1
6			01:32.1
7			01:32.1
8			01:32.1
9			01:32.1
10			01:32.1
11			01:32.1
12			01:32.1
13			01:32.1
14			01:32.1
15			01:32.1
16			01:32.1
17			01:32.1
18			01:32.1
19			01:32.1
20			01:32.1

Table 4: Lap Time across 1 Lap vs. 5 Lap vs. 20 Laps

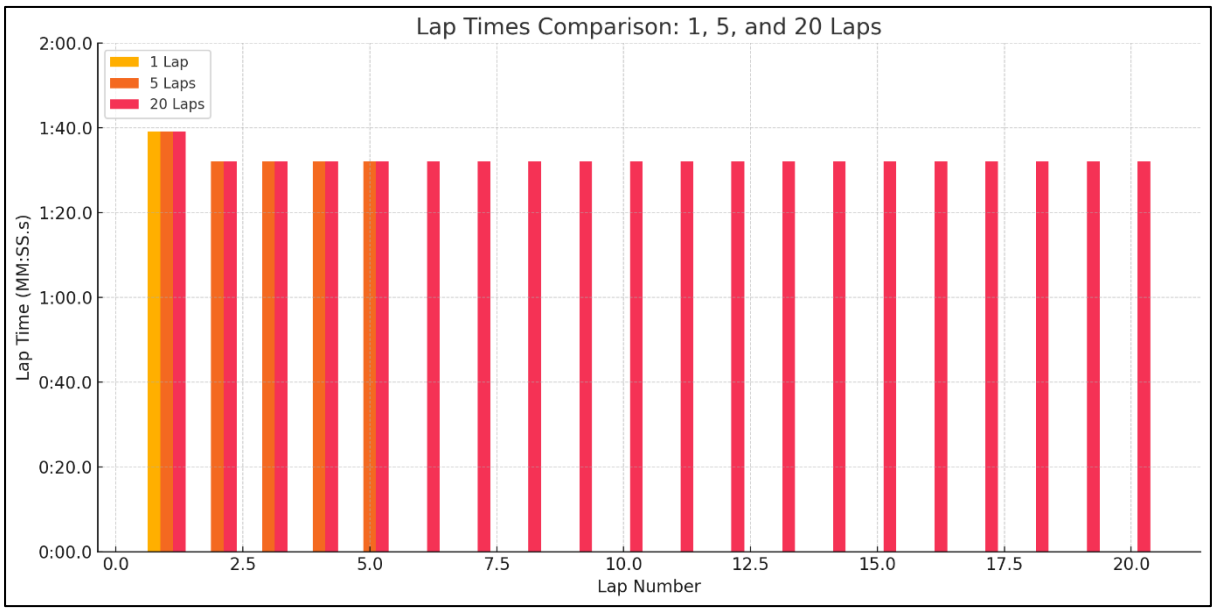


Figure 22: Comparison of Lap times over Lap Counts

14.1.5 Experiment Success Criteria:

- Lap time standard deviation ≤ 0.75% of mean lap time across valid laps.
- Zero critical faults (no collision/spin); minor recoveries permitted if lap finishes.

14.2 Experiment 2 — Development Support (Software Changes and Fine-Tuning for V3M)

Goal: Provide a flexible user interface with interchanging parameters suitable for adaptation of any track's layout

14.2.1 Experiment Scenario / Design:

- A/B Testing with same scenario under identical conditions.
- Multiple tracks testing with fine-tuned parameters settings (3 multiple tracks shown for variability).
- Fine-Tuning Steering Controls: Angular Offset
- Fine-Tuning PIDs: Throttle (Kp, Ki, Kd)
- Fine-Tuning Differential Torque (OFFLINE PURPOSES ONLY)

12.2.2 Experiment Parameters:

Parameter	Description	Default / Range	Notes
Multiple Tracks	Track variability with Sao Paulo, Berlin & Monaco	Default	Same trajectory and velocity profile for all runs.
Velocity Profile	Target speed vehicle must achieve per distance marker	Default	By default set by system
Steer Control Parameters	**Angular Offset**, Lateral Offsets, Individual Lateral Offset Saturations and Steering Control Saturations	**Gains: 40.0**	Set for one unique run on each track
Pedal Control Parameters	Throttle Kp, **Throttle Ki**, Throttle Kd, Future Target Distance (TD)	**Throttle Ki: 4.0**	Set for one unique run on each track
Torque Control Parameters	Max Engine Brake Torque, **Driving Differential Gain**, Max Drive Torque	**Driving Diff Gain: 10.0**	Set for one unique run on each track
Forced Driver Inputs	Steering Angle, Accelerator Pedal, Brake Pressure, MDriveFL, MDriveFR, MDriveRL, MDriveRR, MBrakeFL, MBrakeFR	Default	By default set by system
Autonomy Mode	Race Car Control setting	Full V3M autonomy	No human intervention
Lap Count	Number of consecutive laps	Min: 1, 5 and Max: ≥ 20 laps	First lap is out lap
Sim: Environment	Other cars or obstacles	Off	Avoid external variability
Logging Rate	Sampling frequency	≥ 100 Hz states	Ensure disk throughput is sufficient
Incident Auto-Reset	Auto abort after incident	Enabled	Resume next lap without manual input
Acceptance Thresholds	Targets to judge success	V3M stability through corners, velocity error	Used for evaluation; not a runtime input

	reduction, total controllability

Table 5: Parameters used for Experiment 2

14.2.3 Experiment Procedure/Methodology:

1. Run multiple tracks such Sao Paulo, Berlin and Monaco by changing one of the parameters
2. Deploy New parameters setups; rerun with same default seeds and profiles.
3. Execute perturbations; repeat runs; collect steering, padel and torque controls and counts and responsiveness.

14.2.4 Experiment Results:

Angular Offset Observations:

Due to lack of complexity in track layouts of Sao Paulo and Berlin, the V3M was able to navigate the reference line with ease. In the case of Monaco, the vehicle spins out following a high-speed corner before a tight turn. Tuning the Angular Offset to the value of 40.0, helped the V3M become more responsive in this scenario, and successfully navigates the remaining 3 laps.

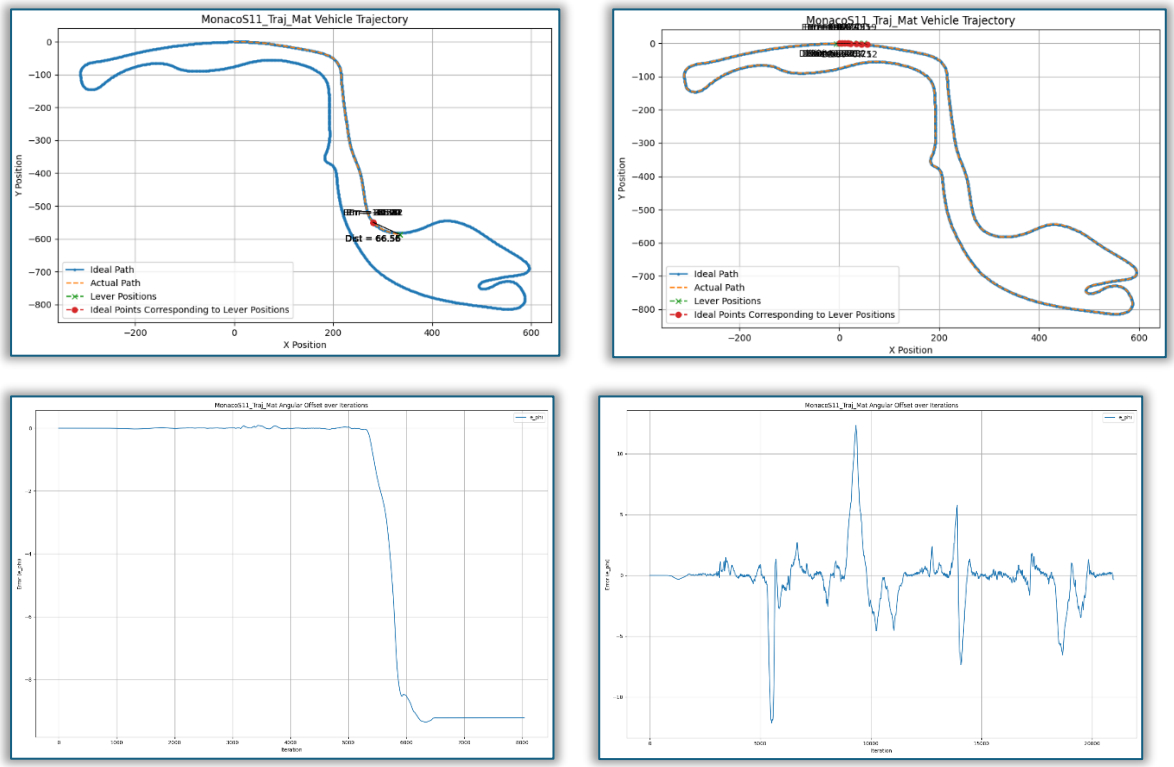


Figure 23: Default Angular Offset vs 40.0 Angular Offset at Monaco

Throttle Ki Observations:

Integrals with fix steady state errors over time and employing greater values should eliminate errors far more quickly. In the case of the V3M throttle, a higher Throttle Ki should mean more aggressive acceleration to minimise velocity error. This is seen across all tracks tested.

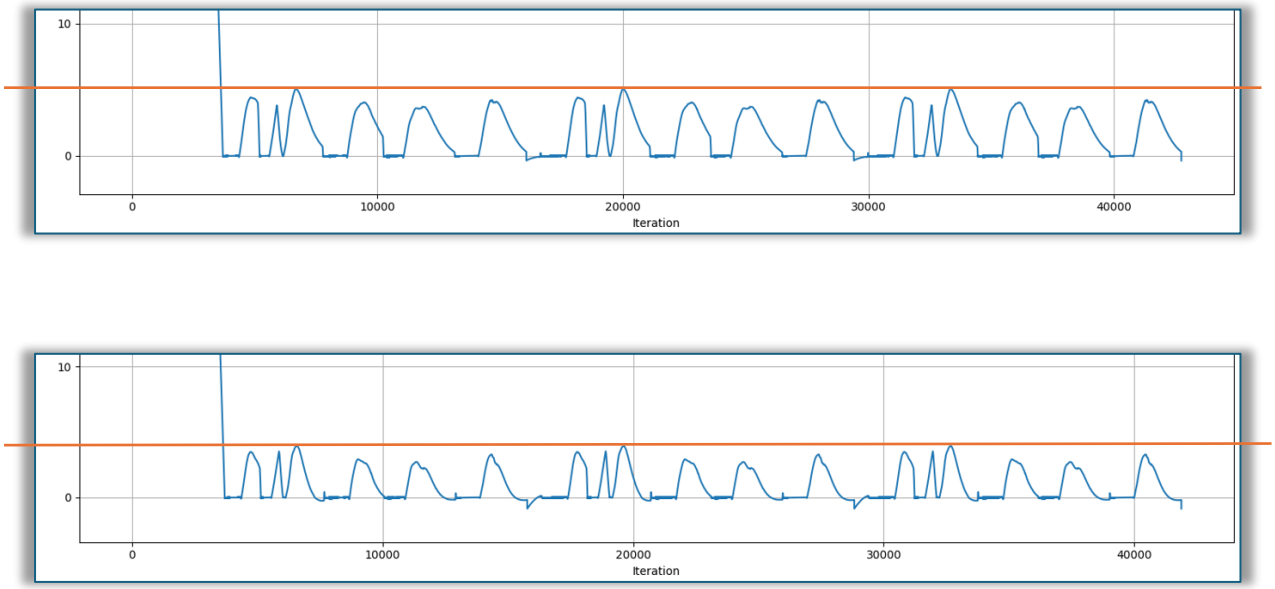


Figure 24: Default Throttle Ki Velocity Error vs 4.0 Ki Velocity Error in Berlin

Default Throttle Ki Lap Times in Berlin	4.0 Throttle Ki Lap Times in Berlin
01:36:777	01:35:068
01:20:027	01:18:305
01:20:022	01:18:305

Table 6: Lap Times for differing Throttle Ki's in Berlin

Driving Differential Gain Observation

The torque control parameters serve to help adjust the torque calculations within offline. Adjusting these values provide flexibility in calculating the specific torques to each wheel during a simulation run. Adjust the Driving Differential Gain value for example, increases the sensitivity of torque entering either wheel based on their current wheel velocity. For each track tested, the torque values were higher than the default values, allowing for full control over the vehicle's differential abilities.

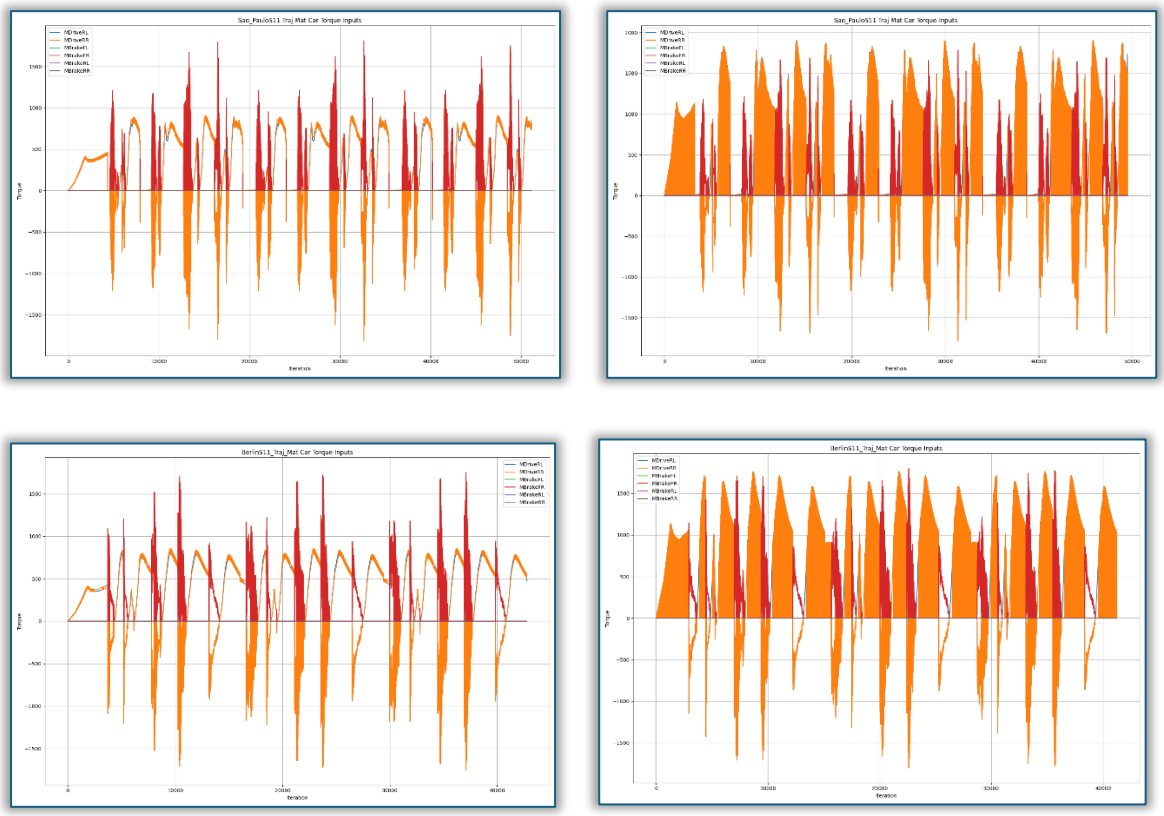


Figure 25: Default Driving Differential Gain Torques vs 10.0 Differential Gain Torques in Sao Paulo and Berlin

14.2.5 Experiment Success Criteria

- Full control over V3M parameters
- Improvement in performance through parameter tweaking
- Improved stability through corners following steering adjustments

14.3 Experiment 3 - Performance Velocity Targeting

Goal: Achieve target velocity adherence while minimizing control aggressiveness and oscillation.

14.3.1 Experiment Scenario / Design:

- Track: Same track used (Berlin) across the experiment to compare velocity %
- Laps: 3 laps (configured with set parameters), no restarts unless safety triggers.
- Mode: Run in Dev Mode and DiL mode.

14.3.2 Experiment Parameters:

Parameter	Description	Default / Range	Notes
Tracks	Only one track used for this experiment: Berlin	Default	Same trajectory across
Velocity Profile	**Target speed vehicle must achieve per distance marker**	**Velocity: 80%, 90% & 100%**	**Set as per**
Steer Control Parameters	Angular Offsets, Lateral Offsets, Individual Lateral Offset Saturations and Steering Control Saturations	Default	By default set by system
Pedal Control Parameters	Throttle Kp, Throttle Ki, Throttle Kd, Future Target Distance (TD)	Default	By default set by system
Torque Control Parameters	Max Engine Brake Torque, Driving Differential Gain, Max Drive Torque	Default	By default set by system
Forced Driver Inputs	Steering Angle, Accelerator Pedal, Brake Pressure, MDriveFL, MDriveFR, MDriveRL, MDriveRR, MBrakeFL, MBrakeFR	Default	By default set by system
Autonomy Mode	Race Car Control setting	Full V3M autonomy	No human intervention
Lap Count	Number of consecutive laps	3 laps	First lap is out lap
Sim: Environment / Traffic	Other cars or obstacles	Off	Avoid external variability
Logging Rate	Sampling frequency	≥ 100 Hz states	Ensure disk throughput is sufficient
Incident Auto-Reset	Auto abort after incident	Enabled	Resume next lap without manual input
Acceptance Thresholds	Targets to judge success	Current velocity reaches target velocity; Zero crash outs	Used for evaluation; not a runtime input

Table 7: Parameters used for Experiment 3

14.3.3 Experiment Procedure/ Methodology:

- Execute tuning matrix, log velocity error and control signals.
- Select Pareto-optimal settings for (RMS velocity error, control effort).

14.3.4 Experiment Results:

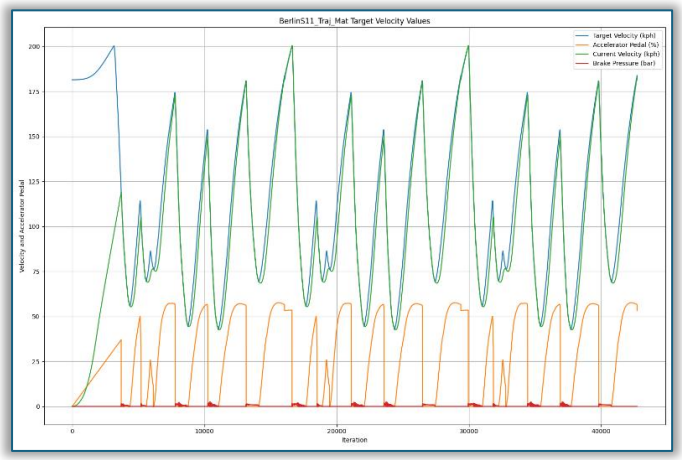


Figure 26: Speed trace for 80% Velocity Run

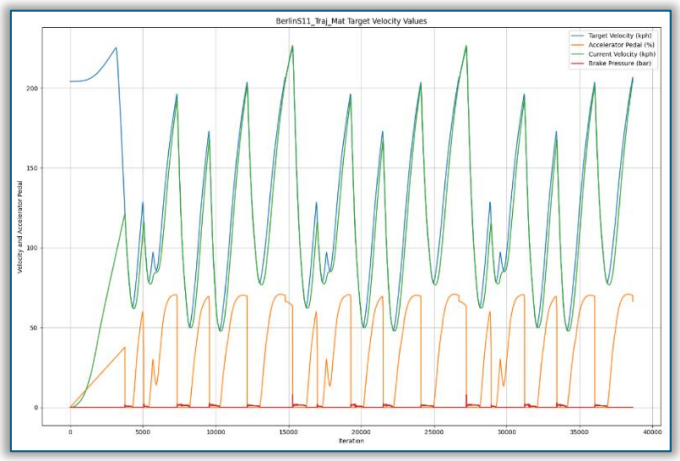


Figure 27: Speed Trace for 90% Velocity Run

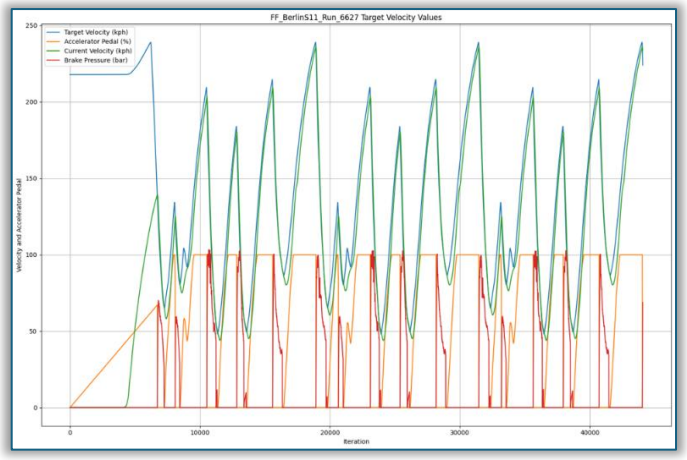


Figure 28: Speed Trace for 100% Velocity Run (Maximum Grip)

80% Target Velocity Lap Time	90% Target Velocity Lap Time	100% Target Velocity Lap Time
1:20:022	1:11:627	1:01:980

Table 8: Consistent Lap Times achieved against each target velocity

14.3.5 Experiment Success Criteria

- Current velocity in Figures 21, 22, and 23 able to match velocity for each target percentage
- V3M is able to complete 3 laps without spinning out reaching high performance
- Lap times reflect each target velocity threshold achieved

14.4 Experiment 4 — Flexible Control & Override Blending

Goal: Validate the seamless integration of forced driver inputs by V3M user

14.4.1 Experiment Scenario / Design:

- Track: Same track used across to test autonomous driving and consistency of laps
- Laps: 1 Out lap, error handling disabled to allow forced inputs
- Mode: Run in Dev Mode and DiL mode.

14.4.2 Experiment Procedure / Methodology:

- Allow V3M to operate normally for one minute of the lap
- After one minute of the lap, force one input to client (throttle, brake, steering input)
- Run multiple out laps with different forced driver inputs at exact same time marker and evaluate outcome

14.4.3 Experiment Parameters:

Parameter	Description	Default / Range	Notes
Tracks	Only one track used for this experiment: Berlin	Default	Same trajectory for all runs.
Velocity Profile	Target speed vehicle must achieve per distance marker	Default	By default set by system
Steer Control Parameters	Angular Offsets, Lateral Offsets, Individual Lateral Offset Saturations and Steering Control Saturations	Default	By default set by system
Pedal Control Parameters	Throttle Kp, Throttle Ki, Throttle Kd, Future Target Distance (TD)	Default	By default set by system
Torque Control Parameters	Max Engine Brake Torque, Driving Differential Gain, Max Drive Torque	Default	By default set by system
Forced Driver Inputs	**Steering Angle**, Accelerator Pedal, Brake Pressure, MDriveFL, MDriveFR, MDriveRL, MDriveRR, MBrakeFL, MBrakeFR	**Steering Angle: 5** Accelerator Pedal/Drive Torque: 0 Brake/Brake Torque: 0	Forcing a custom steering input compared to uninterrupted simulation
Autonomy Mode	Race Car Control setting	Partial V3M autonomy	Human intervention after one minute of driving
Lap Count	Number of consecutive laps	1 lap	Consistent time of intervention on all laps
Sim: Environment / Traffic	Other cars or obstacles	Off	Avoid external variability
Logging Rate	Sampling frequency	≥ 100 Hz states	Ensure disk throughput is sufficient
Incident Auto-Reset	Auto abort after incident	Disabled	Allows error after custom driver inputs are forced
Acceptance Thresholds	Targets to judge success	Deviation from target reference line, forced driver input displayed in respective output graph	Used for evaluation; not a runtime input

Table 9: Parameters used for Experiment 4

14.4.4 Experiment Results:

Forced Driver Inputs

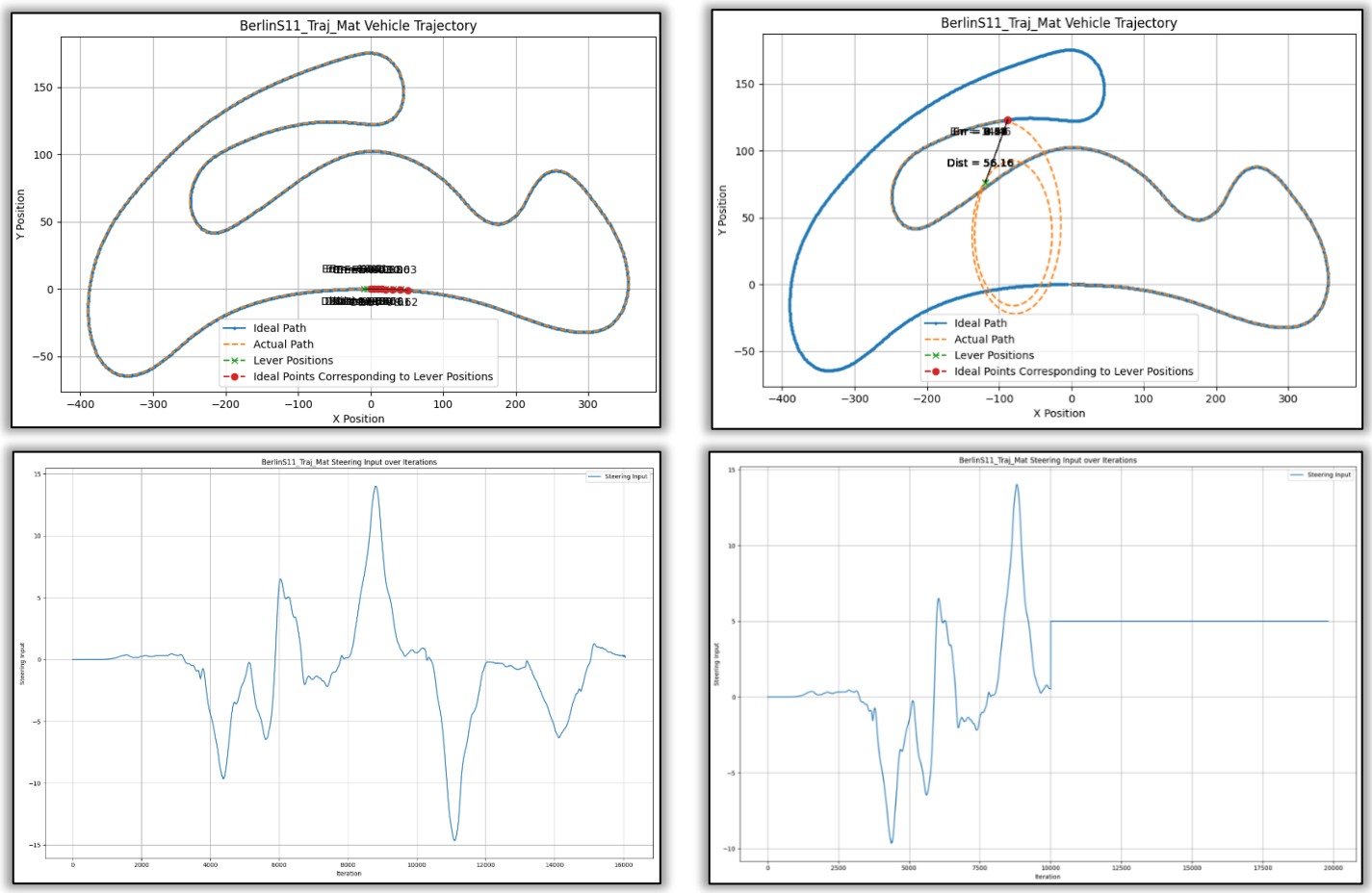


Figure 29: Default Driver Input (Left), Forced Driver Input (Right)

The forced driver inputs experiment for this section only demonstrates forcing a steering angle input. For further analyses, please further use Accelerator Pedal/Driveshaft torques and Brake Pressure/Brake torques parameters and rerun experiment in similar way to the above

14.4.5 Experiment Success Criteria

- Vehicle stops following the track reference line in accordance with the forced driver input submitted
- Steering input logged as forced driver input (see Figure 28)

14.5 Benefits of the V3M

The V3M delivers a range of operational and performance benefits across control systems development, simulation workflows, and track preparation. Its integration into the Driver-in-the-Loop simulator enables faster development cycles, consistent testing, and expanded test coverage:

- Unmanned Simulation Capability

Enables simulation runs without a physical driver present, allowing 24/7 testing and freeing simulator time for additional projects.

- Unlimited Lap Execution

Supports continuous driving over unlimited laps, ideal for endurance testing, system validation, and long-term performance analysis.

- Consistent, Repeatable Inputs

Provides deterministic driver behaviour, ensuring reliable comparisons across VCM software versions, calibration sets, and vehicle setups.

- Throttle and Brake Forcing Capability

Allows injection of forced driver inputs (e.g., throttle surges, braking overrides) for hybrid testing or system stress scenarios.

- Driver Behaviour Tuning

Adjustable model parameters allow simulation of different driving styles and skill levels, supporting a wider range of test cases and performance profiles.

- Accelerated VCM Development

Enables faster iteration on control algorithms by providing a stable and responsive test driver that immediately reflects software changes.

- Improved Test Efficiency

Reduces setup and execution overhead by streamlining the simulation process and eliminating the variability introduced by human drivers.

- High-Fidelity Lap Simulation

Achieves up to 100% target velocity on high-grip track conditions, providing realistic performance benchmarks and lap time targets.

15. Integration with Jaguar TCS Racing Simulator

The V3M is designed for flexible integration with multiple stages of the simulation environment—ranging from offline vehicle models to full Driver-in-the-Loop simulators. This modular approach ensures safe, scalable development and validation of control logic before engaging with high-fidelity hardware.

15.1 Integration Phases

Phase 1: Offline Integration using the Vehicle Model API

- **Objective**: Validate V3M logic and control strategies in a sandboxed, fully controllable environment.
- **Process**:
  - Use a lightweight Python-based vehicle dynamics API to simulate vehicle behaviour.
  - The V3M calculates control outputs (e.g., steering, driveshaft torques, brake torques).
  - Simulated responses (e.g., position, velocity, orientation) are fed back to refine control behaviour.
- **Benefits**:
  - High-speed iteration.
  - Enables early development of path planning, target tracking, and controller tuning.

Phase 2: Emulator UI Integration

- **Objective**: Connect to a mid-fidelity graphical emulator that mirrors the DiL simulator interface without engaging physical hardware.
- **Process**:
  - The V3M runs in parallel with an emulator UI.
  - Control outputs are sent to the emulator via a UDP server/client interface (*UdpCommunication - Python Wiki*, 2025)**.**
  - Raw drive and brake torques are calculated by the V3M and transmitted directly.
- **Benefits**:
  - Validates data communication, signal timing, and control logic in a realistic but safe environment.
  - Simulates real-time constraints and latency.
  - Verifies stability and control under near-live conditions.

Phase 3: Full Driver-in-the-Loop (DiL) Simulator Integration

- **Objective**: Integrate the V3M with the full-scale DiL simulator for complete system evaluation.
- **Process**:
  - The UDP client transmits V3M outputs to the DiL control stack.
  - During initial integration, the system used raw torque values (drive torque, brake torque).
  - As integration matured, and the **DiL Vehicle Control Module (VCM) software** was introduced, the V3M transitioned to sending **throttle percentages** and **brake pressures** instead of raw torques.

- This required implementing **PID controllers** to calculate throttle percentages and brake pressures using velocity error against the target velocity profile

- **Benefits**:
  - Allows complete system validation and operation of DiL simulator without the use of a driver
  - Ensures that the V3M inputs through actual realistic driver inputs as throttle and brake pressure

15.2 Integration Summary Diagrams

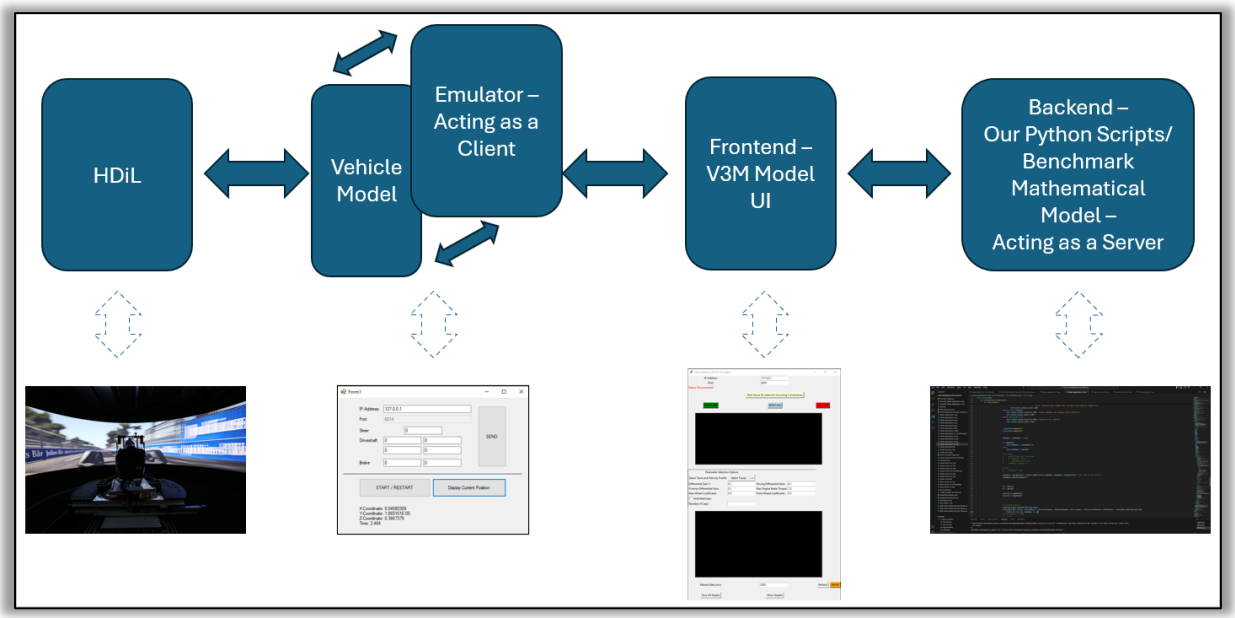


Figure 30: Architecture of UI using the DiL, Vehicle Model/Emulator and the V3M

15.3 User Interface Designs

The V3M application's GUI, which uses tkinter (*Graphical user interfaces with Tk*, 2025), has evolved significantly from its initial release, improving both functionality and user experience to better support development workflows and simulation operations.

Initial Design

- The early GUI was minimalistic, focusing solely on core functionality.
- Offered limited customization and control, with basic start/stop operations.
- Primarily served as a technical frontend for internal testing and scripting.

Progressive Enhancements

As user demands and simulation complexity grew, the GUI was refined with the following improvements:

- **Expanded Control Capabilities**
  - Introduced more customisable parameters force inputs and lap number selection
  - Enabled toggling between manual and autonomous operation.
- **Chronological Workflow Layout**

- The interface was redesigned to reflect the logical simulation flow: connection → configuration → execution → monitoring.
- This reduced user errors and streamlined the operation process.

- **Improved Usability**
  - Added tooltips, grouped settings, and visual status indicators.
  - Simplified onboarding for new users while maintaining control flexibility for advanced users.
- **Live Data Feedback Dashboard**
  - Enabled dynamic tracking of model performance and quick identification of anomalies during runs.
  - Integrated real-time visualization of key metrics such as speed, lap time, lap progression and vehicle position

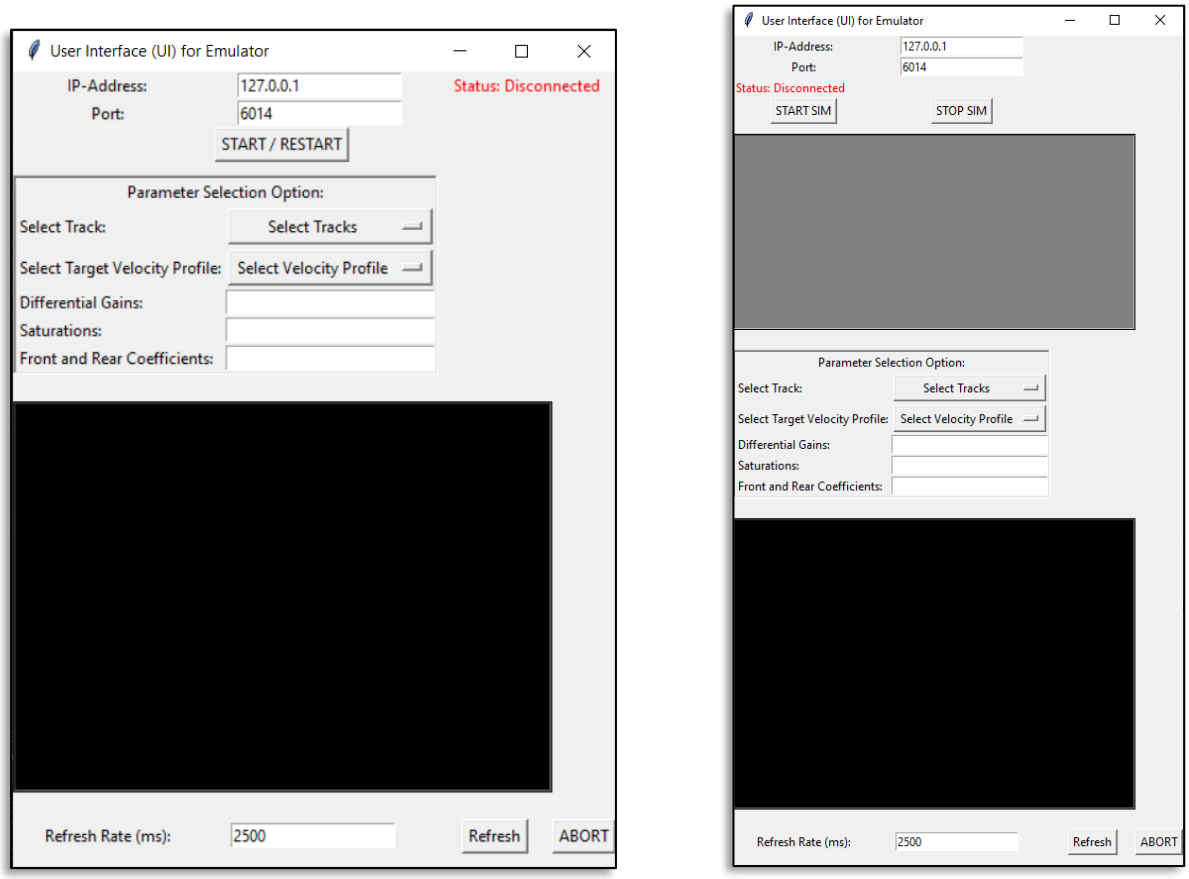


Figure 31: Versions 4 and 5 of the V3M GUI

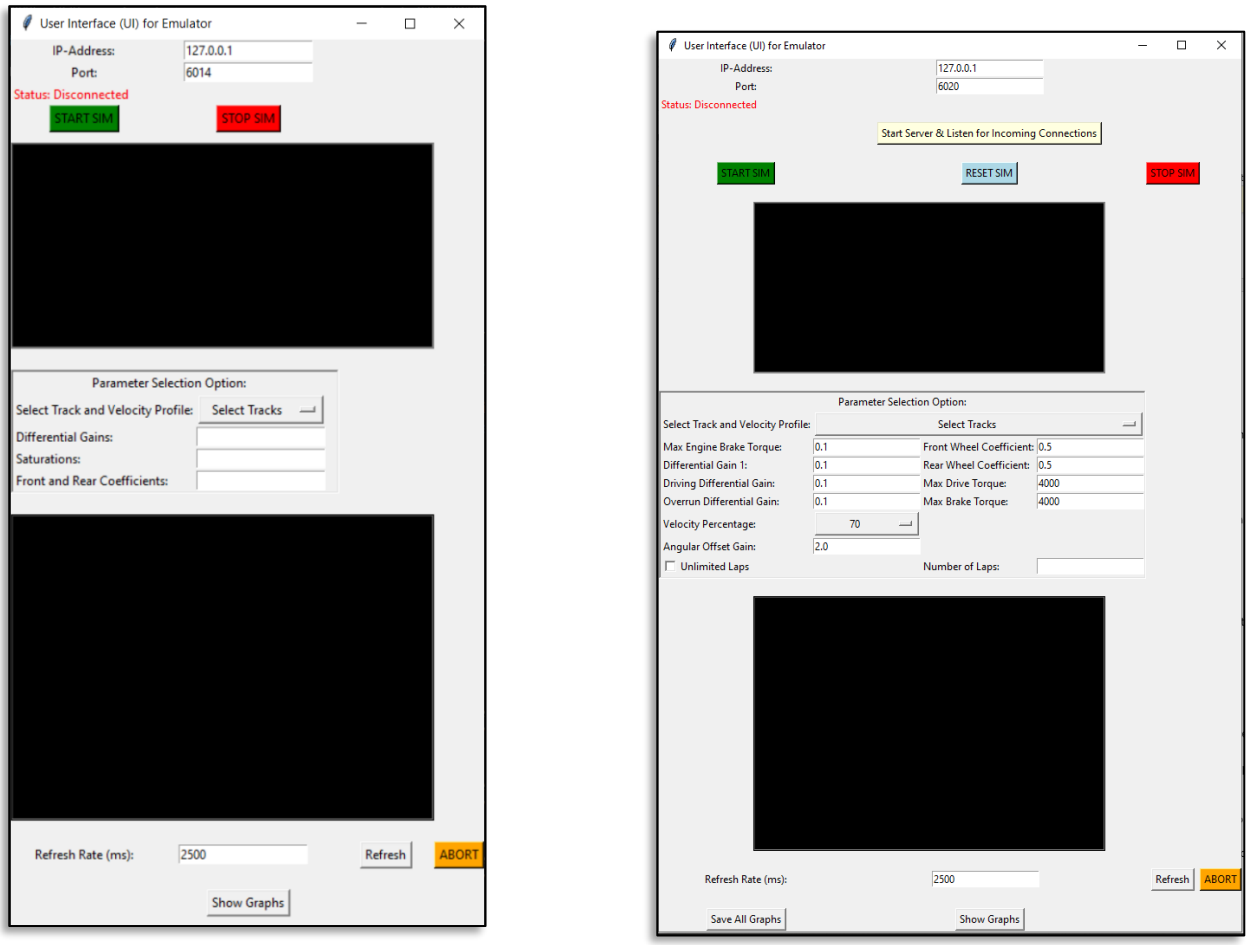


Figure 32: Later Versions 9 and 11 of the V3M GUI

16. Appendix

16.1 Control Parameter Definitions

Driver Model Parameters

- **Select Track File:**
  - Choose a CSV file that defines the reference track to be used during the simulation.
- **Velocity Percentage (%):**
  - Adjust the overall vehicle speed by setting a percentage of the base velocity.
  - (e.g., 100% = full speed)
- **Unlimited Laps:**
  - Enable continuous simulation with no lap limit. The vehicle will loop around the track indefinitely.
- **Number of Laps:**
  - Define the fixed number of laps the simulation will run before stopping.
- **DiL Mode (Driver-in-the-Loop):**
  - Activates real-time user control of the vehicle via input devices for steering, throttle, and braking.
- **Offline Mode (Invert X and Y):**
  - Inverts the X and Y coordinates of the reference line. Useful for debugging or coordinate system adjustments.
- **Auto-Abort:**
  - Automatically stops the simulation if the vehicle deviates outside the defined track boundaries.
- **RESET PARAMETERS:**
  - This button appears in every Parameter Frame. Resets all parameters in designated frame to default values.

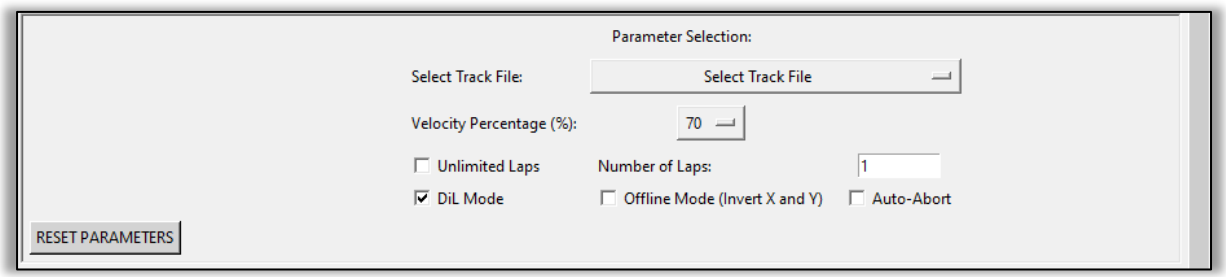


Figure 33: Parameter Selection Window

Steering Control Parameters:

- **Angular Offset Gain:**
  - Gain factor used to correct the vehicle's heading angle relative to the reference path.
- **Lateral Offset Gains:**
  - Gain values applied based on the vehicle's lateral distance from the reference path, adjusted by lookahead distance.
- **Individual Lateral Offset Saturations:**
  - Maximum allowable correction applied for individual lateral offset computations.
- **Total Lateral Offset Saturations:**

- Cumulative output limit from the lateral steering control system to prevent excessive input.

- **Steering Control Saturations:**
  - Overall saturation for steering input

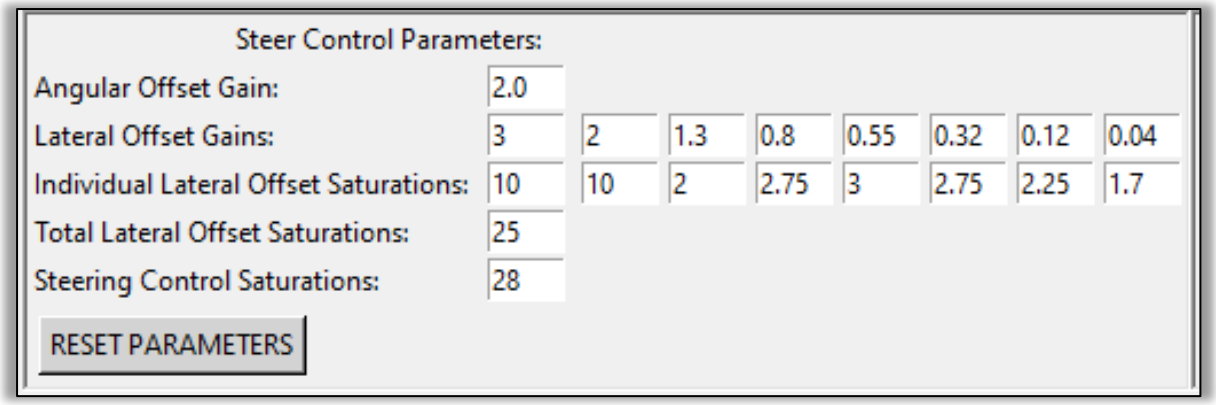
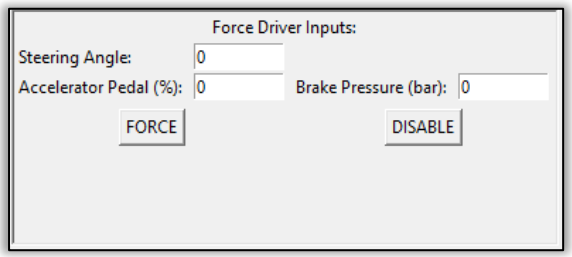


Figure 34: Steer Control Parameters Window

Force Driver Inputs:

- **Steering Angle:**
  - Manually override the steering input.
  - (Negative = Left turn, Positive = Right turn)
- **DiL Mode:**
  - **Accelerator Pedal (%):**
    - Force a fixed accelerator pedal position.
    - (Range: 0–100%)
  - **Brake Pressure (bar):**
    - Set a specific brake pressure input in bars.
- **Offline Mode:**
  - **MDriveFL / MDriveFR / MDriveRL / MDriveRR:**
    - Set drive torque values for each wheel independently (Front Left, Front Right, Rear Left, Rear Right).
  - **MBrakeFL / MBrakeFR:**
    - Set brake torque values for the front wheels (Left and Right).



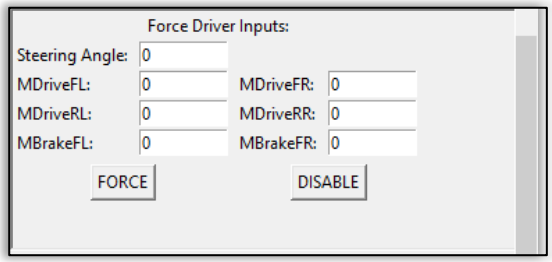


Figure 35: Force Driver Inputs Window (DiL and Offline Modes)

Throttle and Brake Control (PID Controller)

- **Throttle Kp / Ki / Kd:**
  - Set the proportional (Kp), integral (Ki), and derivative (Kd) gains for throttle control.
- **Brake Kp / Ki / Kd:**
  - Set the proportional (Kp), integral (Ki), and derivative (Kd) gains for brake control.
- **Future TD:**
  - Specify a lookahead distance (in meters) used in feedforward throttle prediction logic.
- **Max Brake Pressure:**
  - Controls the maximum amount of brake pressure sent to the DiL.

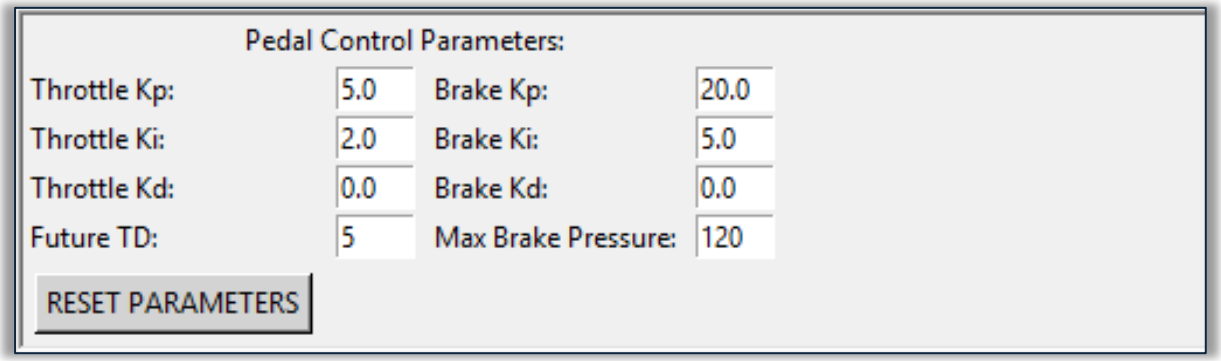


Figure 36: Pedal Control Parameters Window

Torque and Differential Settings

- **Max Engine Brake Torque:**
  - Defines the maximum torque applied by the engine braking system.
  - (Offline testing only)
- **Differential Gain 1:**
  - Gain for balancing torque between the rear wheels under standard conditions.
  - (Offline testing only)
- **Driving Differential Gain:**
  - Adjusts torque distribution between rear wheels during acceleration.
  - (Offline testing only)
- **Overrun Differential Gain:**
  - Controls differential behaviour and torque split during deceleration.
  - (Offline testing only)
- **Max Drive Torque:**
  - Sets the maximum drive torque that can be delivered through the V3M system.
- **Max Brake Torque:**
  - Sets the maximum brake torque that can be applied by the V3M system.

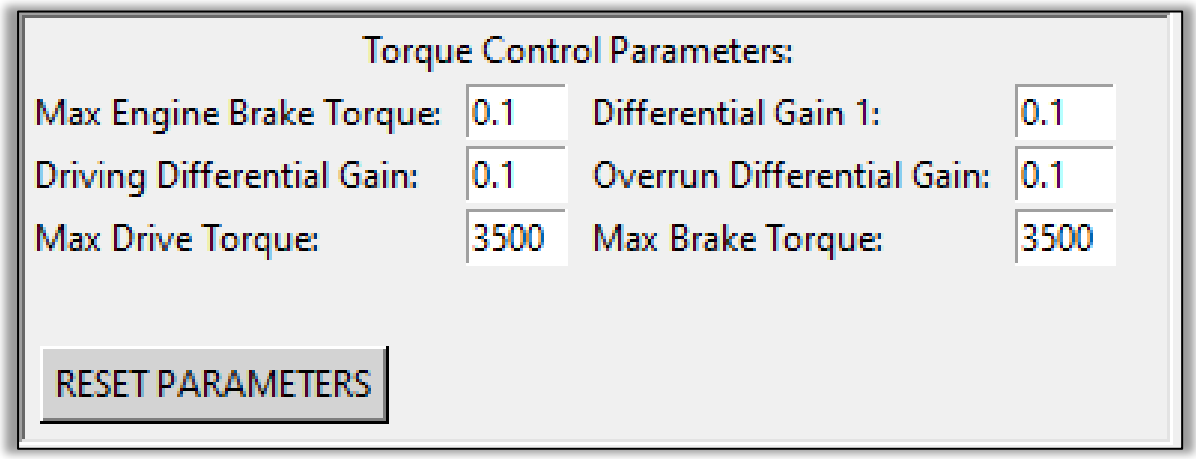


Figure 37: Torque Control Parameters Window

16. 2 Known Errors

- Error starting server: [WinError 10038] An operation was attempted on something that is not a socket.
    - This error will pop up if you start the server and then abort the server without connecting. If this appears, relaunch the server and reconnect. This error will not affect any future runs
- Restarting the V3M after forcing inputs:
    - If the vehicle is not on the reference or far from it after forcing inputs, the V3M may not be able to find the reference line causing the vehicle to spin.
- Connecting to the Offline_Balance_Client:
    - Please ensure you have run all 6 prerequisite .exe programs required by balance, contained here: *L:\Software\FEBalance\Prerequisites\0_FlatFolderv2*
    - Go to the last Build version of balance and install it, located here: *L:\Software\FEBalance\BUILDS\Installer\Build 185*
    - Please also ensure you have a usable Setup ID to access *OpenSetupWarehouse()*, if you receive this error, please contact a DiL engineer and request one
- For further errors, please consult the simulation log for further details

16.3 FAQs

The simulator is not connecting.

- Ensure the simulator is running and configured to connect to IP x.x.x.x on port xxxx.
- Check firewall settings and confirm both applications are on the same network.
- Confirm you pressed **Start Server & Listen** *before* starting the simulator.

The Start button is disabled.

- The simulator must be connected before the **Start** button becomes active.
- Check console logs for any connection errors.

Simulation results are missing, or plots do not appear.

- Check the console for error messages.
- Try running a shorter or simpler test with verified input files.

Application crashes on launch.

- Confirm your system meets the minimum requirements.
- Try running the .exe as administrator.

The Vehicle unexpectedly veers off the driving line or spins out

- Always set the vehicle's starting position to match the coordinates provided in the input data. An incorrect z-coordinate, for example, may cause unintended changes in the vehicle's yaw angle upon descent or after being dropped from a height.
- Possibly due to restarting the V3M after forcing custom inputs into the DiL
- If the issue persists, please manually override the DiL, have use a known operator/driver to drive the DiL for a minute or two and try again

Can I change the V3M parameters during the simulation?

- Only the Velocity Percentage is able to be modified during a simulation run
- The V3M must be reset in order to tweak any parameters during the simulation
- Please press RESET/ABORT before making any changes

Reference List:

Casanova, D. (2000) 'On minimum time vehicle manoeuvring: the theoretical optimal lap'. Available at: http://hdl.handle.net/1826/1091 (Accessed: 21 August 2025).

Discrete-Event Simulation - MATLAB & Simulink (2025). Available at: https://uk.mathworks.com/help/simulink/discrete-event-simulation.html (Accessed: 21 August 2025).

Graphical user interfaces with Tk (2025) *Python documentation*. Available at: https://docs.python.org/3/library/tk.html (Accessed: 21 August 2025).

https://www.fia.com/regulation/category/109 (2025) *Federation Internationale de l'Automobile*. Available at: https://www.fia.com/regulation/category/109 (Accessed: 21 August 2025).

Jaguar TCS Racing Season 11 | Formula E Team (2025) *Jaguar*. Available at: https://www.jaguar.co.uk/jaguar-racing/index.html (Accessed: 21 August 2025).

Matplotlib documentation — Matplotlib 3.10.5 documentation (2025). Available at: https://matplotlib.org/stable/index.html (Accessed: 22 August 2025).

Overview - ATLAS Product Suite (2025). Available at: https://atlas.mclarenapplied.com/introduction/ (Accessed: 22 August 2025).

The Official Home of the ABB FIA Formula E World Championship (2025). Available at: https://www.fiaformulae.com/en (Accessed: 21 August 2025).

The PID Controller & Theory Explained (2025). Available at: https://www.ni.com/en/shop/labview/pid-theory-explained.html (Accessed: 21 August 2025).

UdpCommunication - Python Wiki (2025). Available at: https://wiki.python.org/moin/UdpCommunication (Accessed: 21 August 2025).